Helicity-preserving,pressure-robust MAC-Yee scheme for the Navier-Stokes equation

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Abstract: ,
This paper presents a helicity-preserving and pressure-robust dual-field MAC-Yee scheme for the incompressible Navier–Stokes equations. We first establish the equivalence between the original Navier–Stokes equations and a dual-field formulation, where two sets of velocity and vorticity fields are introduced and shown to coincide under appropriate regularity and boundary conditions. This theoretical result provides a solid foundation for the subsequent numerical discretization. The proposed method is built upon this dual-field weak formulation and discretized on a staggered Cartesian grid, mimicking the structure of the discrete de Rham complex. By applying the compatibility of discrete gradient, curl, and divergence operators, the scheme preserves key physical invariants, including kinetic energy and helicity, at the discrete level. Moreover, the formulation ensures pressure-robustness by decoupling the pressure from the velocity error estimates, thereby avoiding pollution effects in the presence of high pressure gradients. Semi-discrete error estimates are derived under minimal regularity assumptions, demonstrating optimal convergence rates in natural norms. To complement the theoretical analysis, a series of numerical experiments are conducted to validate the convergence behavior, conservation properties, and pressure-robustness of the fully discrete scheme. The results confirm that the method achieves high fidelity in long-time simulations and maintains structural fidelity to the underlying continuum equations.
Keywords: helicity-preserving; Navier-Stokes; marker and cell; pressure-robust

1 Introduction

In this paper, we consider the following three-dimensions incompressible Navier-Stokes equations on a bounded or periodic domain Ω with Lipschitz boundary, and let $T > 0$ be the final time. A dimensionless formulation of this model is expressed as follows:

$$\begin{aligned} \partial_t \boldsymbol{u} + \boldsymbol{\omega} \times \boldsymbol{u} + \frac{1}{Re} \nabla \times \boldsymbol{\omega} + \nabla P &= \boldsymbol{f}, \\ \nabla \times \boldsymbol{u} &= \boldsymbol{\omega}, \\ \nabla \cdot \boldsymbol{u} &= 0. \end{aligned} \tag{1.1}$$

Here, $\boldsymbol{u}$ denotes the velocity filed, $\boldsymbol{\omega}$ denotes the vorticity field, p represents the pressure and Re is the Reynolds number.

The boundary conditions are set as periodic boundary condition or the following two types

$$\boldsymbol{u} \times \boldsymbol{n} = 0,\ P = 0 \text{ on } (0,T) \times \Omega \tag{1.2}$$

$$\text{or } \boldsymbol{u} \cdot \boldsymbol{n} = 0,\ (\nabla \times \boldsymbol{u}) \times \boldsymbol{n} = 0 \text{ on } (0,T) \times \Omega. \tag{1.3}$$

Among the key invariants of the inviscid Navier–Stokes equations are kinetic energy $\mathcal{K} = \int_\Omega \frac{1}{2}\boldsymbol{u}\cdot\boldsymbol{u}\,\mathrm{d}\mathbf{x}$ and helicity $\mathcal{H} = \int_\Omega \boldsymbol{u}\cdot\boldsymbol{\omega}\,\mathrm{d}\mathbf{x}$. Kinetic energy measures the macroscopic motion of the fluidthe and helicity is a measure of the linkage of vortex lines and a topological invariant in three dimensions [18, 6]. In the inviscid limit, i.e., the Euler equations, both kinetic energy and helicity are conserved quantities[44]. These conservation laws are not merely incidental; they reflect the underlying Hamiltonian structure of the Euler equations, where kinetic energy serves as the Hamiltonian functional and helicity emerges as a Casimir invariant associated with the nontrivial kernel of the Poisson operator [31, 32, 39, 30].

The preservation of these invariants at the discrete level is of paramount importance for numerical simulations. Numerical schemes that fail to conserve kinetic energy may introduce spurious dissipation or production, leading to incorrect energy cascade dynamics, particularly in turbulent flows [37, 43, 28, 27, 11]. Structure-preserving discretizations that mimic these conservation laws have been shown to enhance long-time stability, improve physical fidelity in under-resolved simulations, and provide more reliable predictions of complex flow phenomena [12, 35, 3]. This has motivated a growing body of research on numerical methods that respect the geometric and physical structure of the underlying equations [34, 44, 35, 36, 26, 19].

The development of structure-preserving methods for incompressible flows has a long history, rooted in the staggered-grid ideas of Harlow and Welch [16] and the energy-conserving spatial discretizations of Arakawa [1, 2]. These early works already recognised that placing variables at different mesh locations (vertices, edges, faces, cell centres) can mimic the topological action of differential operators and thereby preserve discrete analogues of mass, momentum and energy. Building on this insight, more systematic frameworks emerged, such as mimetic finite differences (MFD) [20, 23, 8], discrete de Rham (DDR) complexes [4, 5] and finite element exterior calculus (FEEC) [4, 5, 7, 17]. All these approaches construct discrete gradient, curl and divergence operators that satisfy the same algebraic identities as their continuous counterparts, guaranteeing that the resulting schemes inherit essential conservation laws and stability properties.

A central theme in incompressible flow simulations is the discrete conservation of kinetic energy and helicity. Classical energy-conserving schemes, often based on skew-symmetric forms of the convective term, have been extensively developed and shown to improve long-time accuracy in turbulent computations [29, 42, 15, 9]. More recently, attention has turned to helicity conservation, with schemes that preserve both energy and helicity proposed by Rebholz [38] and Capuano et al. [10, 41], demonstrating superior fidelity in capturing cascade dynamics.

A particularly promising direction is the use of dual-field formulations [44, 33], where two evolution equations for the velocity are solved in complementary function spaces – typically $H(\mathrm{curl};\Omega)$ and $H(\mathrm{div};\Omega)$. This approach naturally reflects the fact that the continuous velocity belongs to

$H(\mathrm{curl};\Omega)\cap H(\mathrm{div};\Omega)$, a space that is difficult to discretise directly. The dual-field idea was first introduced for two-dimensional incompressible flows by Palha and Gerritsma [33], who developed a mass, energy, enstrophy and vorticity conserving (MEEVC) mimetic spectral element scheme. By staggering the two velocity equations in time, their method yields linear, decoupled algebraic systems while preserving all relevant invariants. Zhang et al. [44] extended this concept to three dimensions on periodic domains, proposing a mimetic dual-field discretisation that employs $\boldsymbol{u}_1 \in H(\mathrm{curl};\Omega), \boldsymbol{u}_2 \in H(\mathrm{div};\Omega)$, the corresponding vorticities $\boldsymbol{\omega}_2 \in H(\mathrm{div};\Omega), \boldsymbol{\omega}_1 \in H(\mathrm{curl};\Omega)$. Combined with a staggered temporal integrator and a discrete de Rham complex, their scheme exactly conserves mass, kinetic energy and helicity in the inviscid limit, and correctly dissipates them in the viscous case. Mao et al. [26] extended the dual field formulations to the incompressible MHD equations to construct a multi physical invariants preserving method.

Despite the promising numerical results achieved by existing dual-field methods, their theoretical foundation remains largely underdeveloped. In particular, the equivalence between the dual-field formulation and the original Navier–Stokes equations has never been rigorously established. Moreover, no error estimates are available for the numerical discretization of such dual-field systems, leaving questions of convergence and accuracy unanswered. The issue of pressure-robustness—whether the velocity error is independent of the pressure approximation—has also not been addressed in the dual-field context, despite its importance for simulations involving high Reynolds numbers and high pressure gradients[21, 24].

In this work, we propose a helicity-preserving and pressure-robust dual-field MAC-Yee scheme for the incompressible Navier–Stokes equations that addresses the theoretical gaps identified above. This formulation naturally accommodates the dual character of the velocity field and enables the construction of a discrete de Rham complex on a staggered Cartesian grid. The spatial discretization follows the philosophy of the MAC scheme for fluid dynamics and Yee's scheme for electrodynamics, assigning degrees of freedom to vertices, edges, faces and cell centers in a way that mimics the topological action of differential operators. Discrete gradient, curl and divergence operators are defined to satisfy the crucial algebraic identities, ensuring that the resulting scheme inherits the exact sequence property of the continuous de Rham complex. For the temporal discretization, we employ a staggered temporal discretiazation that advances the two evolution equations on interleaved time grids, leading to linear and decoupled algebraic systems at each time step while preserving the conservation properties of the semi-discrete formulation.

The main contributions of this paper are summarized as follows:

- Equivalence proof: We rigorously establish the equivalence between the dual-field formulation and the original Navier–Stokes equations under appropriate regularity and boundary conditions. This result provides a solid theoretical foundation for the dual-field approach, showing that the two velocity fields and the two vorticity fields coincide with the physical velocity and vorticity, respectively.

- Pressure-robust semi-discrete error analysis: We derive a pressure-robust error estimates for the semi-discrete scheme in natural norms, demonstrating that the velocity error is independent of the

pressure approximation. This pressure-robustness property ensures that the method performs reliably in the presence of high pressure gradients and at high Reynolds numbers, where conventional methods often suffer from accuracy degradation on these cases.

- Numerical validation: Through a series of numerical experiments, we verify the theoretical convergence rates, the conservation of kinetic energy and helicity, and the pressure-robustness of the proposed method. The results confirm that the scheme achieves high fidelity in long-time simulations and maintains structural fidelity to the underlying continuum equations.

The remainder of this paper is organized as follows. In Section 2, we introduce the continuous dual-field formulation and prove its equivalence to the original Navier–Stokes equations. Section 3 describes the spatial discretization, including the construction of discrete function spaces and operators that form a de Rham complex. In Section 4, we present the semi-discrete scheme and derive pressure-robust error estimates. Section 5 introduces the fully discrete scheme and analyzes its conservation properties. Numerical results are reported in Section 6, and conclusions are drawn in Section 7.

2 Continuous problem

In this section, we rigorously define the continuous dual-field formulation for the incompressible Navier-Stokes equations and establish its theoretical properties. The motivation for this dual-field approach arises from the inherent geometric structure of the velocity field $\boldsymbol{u}$, which simultaneously satisfies the constraints in $H(\mathrm{curl};\Omega)$ and $H(\mathrm{div};\Omega)$. By introducing two distinct velocity representations, $\boldsymbol{u}_1$ and $\boldsymbol{u}_2$, along with their corresponding vorticities $\boldsymbol{\omega}_1$ and $\boldsymbol{\omega}_2$, we can decouple the topological requirements of the flow. This structure mirrors the discrete de Rham complex and ensures that physical invariants—specifically kinetic energy and helicity—are treated consistently.

Then the dual-field weak-formulation is defined as: find $(\boldsymbol{u}_1, \boldsymbol{\omega}_2, P_0) \in H(\mathrm{curl};\Omega) \times H(\mathrm{div};\Omega) \times H^1(\Omega)$ and $(\boldsymbol{u}_2, \boldsymbol{\omega}_1, P_3) \in H(\mathrm{div};\Omega) \times H(\mathrm{curl};\Omega) \times L^2(\Omega)$ such that

$$\left\langle \frac{\partial \boldsymbol{u}_1}{\partial t}, \boldsymbol{\epsilon}_1 \right\rangle_\Omega + \langle \boldsymbol{\omega}_1 \times \boldsymbol{u}_1, \boldsymbol{\epsilon}_1 \rangle_\Omega + \frac{1}{Re} \langle \boldsymbol{\omega}_2, \nabla \times \boldsymbol{\epsilon}_1 \rangle_\Omega + \langle \nabla P_0, \boldsymbol{\epsilon}_1 \rangle_\Omega = \langle \boldsymbol{f}, \boldsymbol{\epsilon}_1 \rangle_\Omega \quad \forall \boldsymbol{\epsilon}_1 \in H(curl;\Omega), \tag{2.1}$$

$$\langle \nabla \times \boldsymbol{u}_1, \boldsymbol{\epsilon}_2 \rangle_\Omega - \langle \boldsymbol{\omega}_2, \boldsymbol{\epsilon}_2 \rangle_\Omega = 0 \quad \forall \boldsymbol{\epsilon}_2 \in H(div;\Omega), \tag{2.2}$$

$$\langle \boldsymbol{u}_1, \nabla \epsilon_0 \rangle_\Omega = 0 \quad \forall \epsilon_0 \in H^1(\Omega), \tag{2.3}$$

$$\left\langle \frac{\partial \boldsymbol{u}_2}{\partial t}, \boldsymbol{\epsilon}_2 \right\rangle_\Omega + \langle \boldsymbol{\omega}_2 \times \boldsymbol{u}_2, \boldsymbol{\epsilon}_2 \rangle_\Omega + \frac{1}{Re} \langle \nabla \times \boldsymbol{\omega}_1, \boldsymbol{\epsilon}_2 \rangle_\Omega - \langle P_3, \nabla \cdot \boldsymbol{\epsilon}_2 \rangle_\Omega = \langle \boldsymbol{f}, \boldsymbol{\epsilon}_2 \rangle_\Omega \quad \forall \boldsymbol{\epsilon}_2 \in H(div;\Omega), \tag{2.4}$$

$$\langle \boldsymbol{u}_2, \nabla \times \boldsymbol{\epsilon}_1 \rangle_\Omega - \langle \boldsymbol{\omega}_1, \boldsymbol{\epsilon}_1 \rangle_\Omega = 0 \quad \forall \boldsymbol{\epsilon}_1 \in H(curl;\Omega), \tag{2.5}$$

$$\langle \nabla \cdot \boldsymbol{u}_2, \epsilon_3 \rangle_\Omega = 0 \quad \forall \epsilon_3 \in L^2(\Omega). \tag{2.6}$$

The strong form of the dual field system can be expressed as follows.

$$\partial_t \boldsymbol{u}_1 + \boldsymbol{\omega}_1 \times \boldsymbol{u}_1 + \frac{1}{Re}\nabla \times \boldsymbol{\omega}_2 + \nabla P_0 = \boldsymbol{f}, \tag{2.7}$$

$$\nabla \times \boldsymbol{u}_1 - \boldsymbol{\omega}_2 = 0, \tag{2.8}$$

$$\nabla \cdot \boldsymbol{u}_1 = 0, \tag{2.9}$$

$$\partial_t \boldsymbol{u}_2 + \boldsymbol{\omega}_2 \times \boldsymbol{u}_2 + \frac{1}{Re}\nabla \times \boldsymbol{\omega}_1 + \nabla P_3 = \boldsymbol{f}, \tag{2.10}$$

$$\nabla \times \boldsymbol{u}_2 - \boldsymbol{\omega}_1 = 0, \tag{2.11}$$

$$\nabla \cdot \boldsymbol{u}_2 = 0. \tag{2.12}$$

Now we will prove the equivalence between the origin system (1.1) and the dual-system (2.1)-(2.6).

Theorem 2.1. *Suppose that the solutions of dual-field weak-formulation (2.1)-(2.6) also satisfy the strong form (2.7)-(2.12), then* $\boldsymbol{u} = \boldsymbol{u}_1 = \boldsymbol{u}_2$, $\boldsymbol{\omega} = \boldsymbol{\omega}_1 = \boldsymbol{\omega}_2$ *and* $\nabla P = \nabla P_0 = \nabla P_3$.

Proof. Firstly, since the solutions $(\boldsymbol{u}_1, \boldsymbol{u}_2, \boldsymbol{\omega}_1, \boldsymbol{\omega}_2, P_0, P_3)$ satisfy the strong form, we have

$$\begin{aligned}
&\boldsymbol{u}_1, \boldsymbol{u}_2 \in L^2(0,T;\boldsymbol{H}^2(\Omega)) \cap H^1(0,T;\boldsymbol{L}^2(\Omega)),\\
&\boldsymbol{\omega}_1, \boldsymbol{\omega}_2 \in L^2(0,T;\boldsymbol{H}^1(\Omega)) \cap H^1(0,T;\boldsymbol{H}^{-1}(\Omega)),\\
&P_0, P_3 \in L^2(0,T;H^1(\Omega)).
\end{aligned}$$

The above regular conditions imply that $\boldsymbol{u}_1, \boldsymbol{u}_2 \in L^2(0,T;H(\mathrm{div};\Omega)) \cap L^2(0,T;H(\mathrm{curl};\Omega))$.
Secondly, we will prove that the solutions of (1.1) satisfy the equations (2.1)-(2.6). Let $(\boldsymbol{u}, \boldsymbol{\omega}, P)$ be the solutions of (1.1), then we define a group of fields $\boldsymbol{u}_1 = \boldsymbol{u}_2 = \boldsymbol{u}$, $\boldsymbol{\omega}_1 = \boldsymbol{\omega}_2 = \boldsymbol{\omega}$, $P_0 = P_3 = P$. According to the boundary condition (1.3), we have $\boldsymbol{u}_1 \cdot \boldsymbol{n} = 0$ and $\boldsymbol{\omega}_1 \times \boldsymbol{n} = 0$ on $\partial\Omega$.

- For any $\boldsymbol{\epsilon}_1 \in H(\mathrm{curl};\Omega)$, $\langle \nabla \times \boldsymbol{\omega}_2, \boldsymbol{\epsilon}_1 \rangle_\Omega = \langle \boldsymbol{\omega}_2, \nabla \times \boldsymbol{\epsilon}_1 \rangle_\Omega - \int_{\partial\Omega} (\boldsymbol{\omega}_2 \times \boldsymbol{n}) \cdot \boldsymbol{\epsilon}_1 \,\mathrm{d}S$. Therefore we can obtain (2.1).
- Since $\nabla \times \boldsymbol{u}_1 = \nabla \times \boldsymbol{u} = \boldsymbol{\omega} = \boldsymbol{\omega}_2$, (2.2) holds true.
- For any $\epsilon_0 \in H^1(\Omega)$, we have $\langle \nabla \cdot \boldsymbol{u}_1, \boldsymbol{\epsilon}_0 \rangle_\Omega = -\langle \boldsymbol{u}_1, \nabla \epsilon_0 \rangle_\Omega + \int_{\partial\Omega} \epsilon_0 (\boldsymbol{u}_1 \cdot n) \,\mathrm{d}S = 0$, thus we can obtain (2.3).
- Let $\boldsymbol{\epsilon}_2 \in H(\mathrm{div};\Omega)$, $\langle \nabla P_3, \boldsymbol{\epsilon}_2 \rangle_\Omega = -\langle P_3, \nabla \cdot \boldsymbol{\epsilon}_2 \rangle_\Omega + \int_{\partial\Omega} P_3 \boldsymbol{\epsilon}_2 \cdot \boldsymbol{n} \,\mathrm{d}S$. Then we can get (2.4).
- For any $\boldsymbol{\epsilon}_1 \in H(\mathrm{div};\Omega)$, it follows $\langle \nabla \times \boldsymbol{u}_2, \boldsymbol{\epsilon}_1 \rangle_\Omega = \langle \boldsymbol{u}_2, \nabla \times \boldsymbol{\epsilon}_1 \rangle_\Omega - \int_{\partial\Omega} (\boldsymbol{u}_2 \times \boldsymbol{n}) \cdot \boldsymbol{\epsilon}_1 \,\mathrm{d}S = \langle \boldsymbol{u}_2, \nabla \times \boldsymbol{\epsilon}_1 \rangle_\Omega - \int_{\partial\Omega} (\boldsymbol{\epsilon}_1 \times \boldsymbol{n}) \cdot \boldsymbol{u}_2 \,\mathrm{d}S = \langle \boldsymbol{u}_2, \nabla \times \boldsymbol{\epsilon}_1 \rangle_\Omega$. Therefore we can obtain (2.5).
- (2.6) can be obtained easily form that $\boldsymbol{u}$ is divergence-free.

Then we can derive some additional boundary conditions.

- The viscous term of (2.1) can be written as

$$\frac{1}{Re} \langle \boldsymbol{\omega}_2, \nabla \times \boldsymbol{\epsilon}_1 \rangle_\Omega = \frac{1}{Re} \langle \nabla \times \boldsymbol{\omega}_2, \boldsymbol{\epsilon}_1 \rangle_\Omega - \int_{\partial\Omega} (\boldsymbol{\omega}_2 \times \boldsymbol{n}) \cdot \boldsymbol{\epsilon}_1 \,\mathrm{d}S. \tag{2.13}$$

Since the strong form (2.7) holds, we obtain that

$$\left\langle \frac{\partial \boldsymbol{u}_1}{\partial t}, \boldsymbol{\epsilon}_1 \right\rangle_\Omega + \langle \boldsymbol{\omega}_1 \times \boldsymbol{u}_1, \boldsymbol{\epsilon}_1 \rangle_\Omega + \frac{1}{Re} \langle \nabla \times \boldsymbol{\omega}_2, \boldsymbol{\epsilon}_1 \rangle_\Omega + \langle \nabla P_0, \boldsymbol{\epsilon}_1 \rangle_\Omega = \langle \boldsymbol{f}, \boldsymbol{\epsilon}_1 \rangle_\Omega \tag{2.14}$$

Subtracting (2.1) from (2.14), it can be seen that $\int_{\partial\Omega} (\boldsymbol{\omega}_2 \times \boldsymbol{n}) \cdot \boldsymbol{\epsilon}_1 \, \mathrm{d}S = 0$.

- From (2.3) and (2.9), we have $\langle \boldsymbol{u}_1, \nabla \epsilon_0 \rangle_\Omega = -\langle \nabla \cdot \boldsymbol{u}_1, \epsilon_0 \rangle_\Omega + \int_{\partial\Omega} (\boldsymbol{u}_1 \cdot \boldsymbol{n}) \epsilon_0 \, \mathrm{d}S = \int_{\partial\Omega} (\boldsymbol{u}_1 \cdot \boldsymbol{n}) \epsilon_0 \, \mathrm{d}S = 0$.
- By applying integration by parts,

$$\begin{aligned} \langle \boldsymbol{u}_2, \nabla \times \boldsymbol{\epsilon}_1 \rangle_\Omega &= \langle \nabla \times \boldsymbol{u}_2, \boldsymbol{\epsilon}_1 \rangle_\Omega - \int_{\partial\Omega} (\boldsymbol{\epsilon}_1 \times \boldsymbol{n}) \cdot \boldsymbol{u}_2 \, \mathrm{d}S \\ &= \langle \nabla \times \boldsymbol{u}_2, \boldsymbol{\epsilon}_1 \rangle_\Omega + \int_{\partial\Omega} (\boldsymbol{u}_2 \times \boldsymbol{n}) \cdot \boldsymbol{\epsilon}_1 \, \mathrm{d}S. \end{aligned} \tag{2.15}$$

Then by virtue of (2.5) and (2.11), subtracting $\langle \boldsymbol{\omega}_1, \boldsymbol{\epsilon}_1 \rangle_\Omega$ from (2.15), we can obtain that

$$\int_{\partial\Omega} (\boldsymbol{u}_2 \times \boldsymbol{n}) \cdot \boldsymbol{\epsilon}_1 \, \mathrm{d}S = 0. \tag{2.16}$$

We define that $\delta \boldsymbol{u} = \boldsymbol{u}_1 - \boldsymbol{u}_2$, $\delta \boldsymbol{\omega} = \boldsymbol{\omega}_1 - \boldsymbol{\omega}_2$ and $\delta P = P_0 - P_3$. Subtracting (2.7) from (2.10), we have

$$\partial_t \delta \boldsymbol{u} + (\boldsymbol{\omega}_1 \times \boldsymbol{u}_1 - \boldsymbol{\omega}_2 \times \boldsymbol{u}_2) + \frac{1}{Re} \nabla \times \delta \boldsymbol{\omega} + \nabla \delta P = 0.$$

Then we can obtain the following equation

$$\partial_t \delta \boldsymbol{u} + \frac{1}{Re} \nabla \times \delta \boldsymbol{\omega} + \nabla \delta P = (\boldsymbol{\omega}_2 \times \boldsymbol{u}_2 - \boldsymbol{\omega}_1 \times \boldsymbol{u}_1). \tag{2.17}$$

Multiplying (2.17) by $\delta \boldsymbol{u}$, it follows

$$\langle \partial_t \delta \boldsymbol{u}, \delta \boldsymbol{u} \rangle_\Omega + \frac{1}{Re} \langle \nabla \times \nabla \times \delta \boldsymbol{u}, \delta \boldsymbol{u} \rangle_\Omega + \langle \nabla \delta P, \delta \boldsymbol{u} \rangle_\Omega = \langle \boldsymbol{\omega}_2 \times \boldsymbol{u}_2 - \boldsymbol{\omega}_1 \times \boldsymbol{u}_1, \delta \boldsymbol{u} \rangle_\Omega . \tag{2.18}$$

The term $\langle \partial_t \delta \boldsymbol{u}, \delta \boldsymbol{u} \rangle_\Omega$ can be written as $\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)}$. The viscous term $\frac{1}{Re} \langle \nabla \times \nabla \times \delta \boldsymbol{u}, \delta \boldsymbol{u} \rangle_\Omega$ can be written as

$$\begin{aligned} \frac{1}{Re} \langle \nabla \times \nabla \times \delta \boldsymbol{u}, \delta \boldsymbol{u} \rangle_\Omega &= \frac{1}{Re} \langle \nabla \times \delta \boldsymbol{u}, \nabla \times \delta \boldsymbol{u} \rangle_\Omega - \int_{\partial\Omega} ((\nabla \times \delta \boldsymbol{u}) \times \boldsymbol{n}) \cdot \delta \boldsymbol{u} \, \mathrm{d}S \\ &= \frac{1}{Re} \|\nabla \times \delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)} \\ &\quad - \Big(\int_{\partial\Omega} ((\nabla \times \boldsymbol{u}_1) \times \boldsymbol{n}) \cdot \delta \boldsymbol{u} \, \mathrm{d}S - \int_{\partial\Omega} ((\nabla \times \boldsymbol{u}_2) \times \boldsymbol{n}) \cdot \delta \boldsymbol{u} \, \mathrm{d}S \Big) \\ &= \frac{1}{Re} \|\nabla \times \delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)} \\ &\quad - \Big(\int_{\partial\Omega} (\boldsymbol{\omega}_1 \times \boldsymbol{n}) \cdot \delta \boldsymbol{u} \, \mathrm{d}S - \int_{\partial\Omega} (\boldsymbol{\omega}_2 \times \boldsymbol{n}) \cdot \delta \boldsymbol{u} \, \mathrm{d}S \Big) \end{aligned} \tag{2.19}$$

By virtue of the original boundary condition $\boldsymbol{\omega}_2 \times \boldsymbol{n}|_{\partial\Omega} = 0$, the derivation boundary condition $\int_{\partial\Omega} (\boldsymbol{\omega}_1 \times \boldsymbol{n}) \cdot \boldsymbol{\epsilon}_1 \, \mathrm{d}S = 0$ and the assumption $\boldsymbol{u}_1, \boldsymbol{u}_2 \in H(\mathrm{div}; \Omega) \cap H(\mathrm{curl}; \Omega)$, we can obtain that the boundary terms are 0. Thus we have

$$\frac{1}{Re} \langle \nabla \times \nabla \times \delta \boldsymbol{u}, \delta \boldsymbol{u} \rangle_\Omega = \frac{1}{Re} \|\nabla \times \delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)}. \tag{2.20}$$

The term $\langle \nabla \delta P, \delta \boldsymbol{u} \rangle_\Omega$ can be treated as

$$\langle \nabla \delta P, \delta \boldsymbol{u} \rangle_\Omega = \langle \nabla P_0, \boldsymbol{u}_1 \rangle_\Omega - \langle \nabla P_0, \boldsymbol{u}_2 \rangle_\Omega - \langle \nabla P_3, \boldsymbol{u}_1 \rangle_\Omega + \langle \nabla P_3, \boldsymbol{u}_2 \rangle_\Omega \tag{2.21}$$

According to (2.3), $\langle \nabla P_0, \boldsymbol{u}_1 \rangle_\Omega = 0$. By virtue of the strong form $\nabla \cdot \boldsymbol{u}_1 = \nabla \cdot \boldsymbol{u}_2 = 0$, the original and derivation boundary conditions that $\boldsymbol{u}_2 \cdot \boldsymbol{n}|_{\partial\Omega} = 0$, $\int_{\partial\Omega} (\boldsymbol{u}_1 \cdot \boldsymbol{n}) \epsilon_0 = 0$, and the assumption that $P_0, P_3 \in \boldsymbol{H}^1(\Omega)$. The other terms on the right-hand side of (2.21) vanish to 0.
Now (2.18) can be written as

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{1}{Re} \|\nabla \times \delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)} = \langle \boldsymbol{\omega}_2 \times \boldsymbol{u}_2 - \boldsymbol{\omega}_1 \times \boldsymbol{u}_1, \delta \boldsymbol{u} \rangle_\Omega . \tag{2.22}$$

For the nonlinear term $\langle \boldsymbol{\omega}_2 \times \boldsymbol{u}_2 - \boldsymbol{\omega}_1 \times \boldsymbol{u}_1, \delta \boldsymbol{u} \rangle_\Omega$, by applying the Hölder inequalityinequality and Young inequality, we have

$$\begin{aligned}
\langle \boldsymbol{\omega}_2 \times \boldsymbol{u}_2 - \boldsymbol{\omega}_1 \times \boldsymbol{u}_1, \delta \boldsymbol{u} \rangle_\Omega &= \langle \boldsymbol{\omega}_2 \times \boldsymbol{u}_2, \delta \boldsymbol{u} \rangle_\Omega - \langle \boldsymbol{\omega}_1 \times \boldsymbol{u}_2, \delta \boldsymbol{u} \rangle_\Omega \\
&\quad + \langle \boldsymbol{\omega}_1 \times \boldsymbol{u}_2, \delta \boldsymbol{u} \rangle_\Omega - \langle \boldsymbol{\omega}_1 \times \boldsymbol{u}_1, \delta \boldsymbol{u} \rangle_\Omega \\
&= \langle \boldsymbol{u}_2 \times \delta \boldsymbol{\omega}, \delta \boldsymbol{u} \rangle_\Omega + \langle \delta \boldsymbol{u} \times \boldsymbol{\omega}_1, \delta \boldsymbol{u} \rangle_\Omega \\
&= \langle \boldsymbol{u}_2 \times \delta \boldsymbol{\omega}, \delta \boldsymbol{u} \rangle_\Omega \\
&\leq C \|\boldsymbol{u}_2\|_{\boldsymbol{L}^\infty(\Omega)} \|\delta \boldsymbol{\omega}\|_{\boldsymbol{L}^2(\Omega)} \|\delta \boldsymbol{u}\|_{\boldsymbol{L}^2(\Omega)} \\
&\leq \frac{1}{2Re} \|\delta \boldsymbol{\omega}\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{Re \|\boldsymbol{u}_2\|^2_{\boldsymbol{L}^\infty(\Omega)}}{2} \|\delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)}.
\end{aligned} \tag{2.23}$$

Combining (2.22) and (2.23), we can observe that

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{1}{2Re} \|\nabla \times \delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)} \leq C \|\delta \boldsymbol{u}\|^2_{\boldsymbol{L}^2(\Omega)}. \tag{2.24}$$

According to the assumption that $\|\delta \boldsymbol{u}(\cdot, 0)\|_{\boldsymbol{L}^2(\Omega)} = 0$ and Gronwall lemma, we can denote that $\delta \boldsymbol{u} = 0$ and $\delta \boldsymbol{\omega} = 0$, thus it implies $\nabla P_3 = \nabla P_0$ since the strong form (2.7) and (2.10) hold. □

3 Discrete problem

The discrete scheme developed in this section is established by integrating the core principles of the MAC scheme for fluid dynamics and Yee' s scheme for electrodynamics. This framework can be fundamentally interpreted as the lowest-order realization of the discrete de Rham (DDR) complex and mimetic finite difference (MFD) methods on a structured Cartesian grid. By aligning the discrete operators with the topological structure of the grid, this approach naturally facilitates extensions to higher-order mimetic discretizations and more general DDR complexes on polyhedral meshes.

A key feature of this spatial discretization is its "physics-compatible" nature. By carefully defining the degrees of freedom on the staggered grid—namely, vertices, edges, face centers, and cell centers — the scheme satisfies a discrete version of the de Rham commutative diagram . This structural preservation ensures that the resulting numerical algorithm rigorously maintains the conservation of essential physical quantities, including kinetic energy , helicity, and mass (divergence-free constraint), at the discrete level.

Discrete de Rahm complex and spatial discretization

To describe the discrete de Rham complex, we first partition the computational domain Ω into a uniform Cartesian grid consisting of $N_x \times N_y \times N_z$ cubic cells. The mesh sizes in the x, y, and z directions are denoted by $\Delta x, \Delta y$, and Δz, respectively. For the sake of simplicity in the subsequent analysis, we assume a uniform mesh size $h = \Delta x = \Delta y = \Delta z$.

The discrete function spaces V_h^0 to V_h^3 are constructed by assigning degrees of freedom to specific geometric entities of the grid to mimic the topological structure of continuous differential forms. Following the staggered grid setting of the MAC-Yee scheme:

- V_h^3 (Cell Centers Ω_C): Located at centers of cell, indexed by $(i-\frac{1}{2}, j-\frac{1}{2}, k-\frac{1}{2})$, representing scalar quantities such as pressure P_3[cite: 46, 137, 529]. A control volume $T^0_{i-\frac{1}{2},j-\frac{1}{2},k-\frac{1}{2}}$ corresponding to $(i-\frac{1}{2}, j-\frac{1}{2}, k-\frac{1}{2})$.
- V_h^2 (Face Centers Ω_F): Located at the centers of cell interfaces, indexed by $(i, j-\frac{1}{2}, k-\frac{1}{2})$, $(i-\frac{1}{2}, j, k-\frac{1}{2})$, and $(i-\frac{1}{2}, j-\frac{1}{2}, k)$, representing the normal components of vector fields such as velocity $\boldsymbol{u}_1$ and vorticity $\boldsymbol{\omega}_1$.
- V_h^1 (Edge Centers Ω_E): Located at the centers of cell edges, indexed by $(i-\frac{1}{2}, j, k)$, $(i, j-\frac{1}{2}, k)$, and $(i, j, k-\frac{1}{2})$, representing the tangential components of vector fields such as velocity $\boldsymbol{u}_2$ and vorticity $\boldsymbol{\omega}_2$.
- V_h^0 (Vertices Ω_V): Located at the grid nodes, indexed by (i, j, k), representing scalar quantities such as pressure P_0.

A control volume corresponding to an index (i, j, k), where i, j, k are integers or half-integers, is denoted by $T_{i,j,k}$ and defined as $T_{i,j,k} = (x_i - \frac{\Delta x}{2}, x_i + \frac{\Delta x}{2}) \times (y_j - \frac{\Delta y}{2}, y_j + \frac{\Delta y}{2}) \times (z_k - \frac{\Delta z}{2}, z_k + \frac{\Delta z}{2}) \cap \Omega$. Thus, for any discrete function $v \in V_h^i$, it can be interpreted as a piecewise constant field over the collection of control volumes, where its value on each $T_{i,j,k}$ is precisely determined by the discrete degree of freedom located at that specific index. Therefore, the L^2 norm of some $v \in V_h^i$ has a natural definition, and the L^2 inner product is denoted by $\langle \cdot, \cdot \rangle_\Omega$.

The spatial discretization relies on the basic difference operator D_1 in the x-direction, defined as:

$$D_1 f(x, y, z) = \frac{f(x + \Delta x/2, y, z) - f(x - \Delta x/2, y, z)}{\Delta x}.$$

The operators D_2 and D_3 are defined analogously for the y and z directions. Using these, we define the discrete gradient ∇_h, curl $\nabla_h \times$, and divergence $\nabla_h \cdot$ operators that map between the respective spaces V_h^i:

- **Gradient**: $\nabla_h q = (D_1 q, D_2 q, D_3 q)$ on $\begin{cases} V_h^1, \text{ if } q \in V_h^0, \\ V_h^2, \text{ if } q \in V_h^0. \end{cases}$
- **Curl**: $\nabla_h \times \boldsymbol{f} = (D_2 f_3 - D_3 f_2, D_3 f_1 - D_1 f_3, D_1 f_2 - D_2 f_1)$ on $\begin{cases} V_h^1, \text{ if } \boldsymbol{f} \in V_h^2, \\ V_h^2, \text{ if } \boldsymbol{f} \in V_h^1. \end{cases}$

- **Divergence**: $\nabla_h \cdot \boldsymbol{f} = D_1 f_1 + D_2 f_2 + D_3 f_3$ on $\begin{cases} V_h^0, \text{ if } \boldsymbol{f} \in V_h^1, \\ V_h^3, \text{ if } \boldsymbol{f} \in V_h^2. \end{cases}$

The discrete function spaces V_h^i and the corresponding discrete gradient, curl, and divergence operators defined above represent the lowest-order realization of the discrete de Rham (DDR) complex and Mimetic Finite Difference (MFD) methods on a structured Cartesian grid. By aligning these discrete operators with the topological structure of the mesh, the scheme preserves the fundamental properties of the continuous differential operators at the algebraic level. Specifically, the compatibility between the projection operators $\mathcal{R}_i$ and the discrete derivatives ensures the commutativity of the diagram. Consequently, we have the following continuous-discrete de Rham commutative diagram [22]:

$$\begin{tikzcd}
V_h^0 \arrow[r, "\nabla_h(\mathrm{d}_h^0)"] \arrow[d, shift left, "I_0"] & V_h^1 \arrow[r, "\nabla_h\times(\mathrm{d}_h^1)"] \arrow[d, shift left, "I_1"] & V_h^2 \arrow[r, "\nabla_h\cdot(\mathrm{d}_h^2)"] \arrow[d, shift left, "I_2"] & V_h^3 \arrow[d, shift left, "I_3"] \\
H^1(\Omega)(V^0) \arrow[r, "\nabla(\mathrm{d}^0)"] \arrow[u, shift left, "\mathcal{R}_0"] & H(\mathrm{curl};\Omega)(V^1) \arrow[r, "\nabla\times(\mathrm{d}^1)"] \arrow[u, shift left, "\mathcal{R}_1"] & H(\mathrm{div};\Omega)(V^2) \arrow[r, "\nabla\cdot(\mathrm{d}^2)"] \arrow[u, shift left, "\mathcal{R}_2"] & L^2(\Omega)(V^3) \arrow[u, shift left, "\mathcal{R}_3"]
\end{tikzcd}$$

The projection operators $\{\mathcal{R}_i\}_{i=0}^3$ are defined as

- $\mathcal{R}_0$ associates to a continuous function its values at all the vertices of the grid.
- $\mathcal{R}_1$ associates to a vector valued functions the circulations on all the edges.
- $\mathcal{R}_2$ associates to a vector valued function the integrals on all the fluxes through all face.
- $\mathcal{R}_3$ associates to a scalar function its integrals on all the cells of the grid.

The reconstruction operators I_i denotes reconstruct the values on the degrees of freedom to the associated histoplation polynomial(equivalent to the lowest order V^i-confirming finite element polynomial). Since the diagram is commutative, the following identity holds

$$\phi_i^h = \mathcal{R}_i(I_i \phi_i^h),\ \phi_i^h \in V_h^i.$$

We can also show the above discrete de Rahm complex is exact, i.e. $\mathrm{Im}(\mathrm{d}_h^k) = \mathrm{Ker}(\mathrm{d}_h^{k+1})$. The relationship $\mathrm{Im}(\mathrm{d}_h^k) \subset \mathrm{Ker}(\mathrm{d}_h^{k+1})$ is obvious. We only show the reverse relationship. For any $\boldsymbol{v}_i^h \in V_h^i$ and $\mathrm{d}_h^i \boldsymbol{v}_i^h = 0$, we have $I_{i+1} \circ \mathrm{d}_h^i \boldsymbol{v}_i^h = 0$, since the the diagram is commutative, it follows $\mathrm{d}^i I_i \boldsymbol{v}_i^h = 0$. According to the commutative property of the continuous de Rahm complex, there exists $\phi_{i-1} \in V^{i-1}$ such that $\mathrm{d}^{i-1}\phi_{i-1} = I_i \boldsymbol{v}_i^h$, using the commutative property again, we obtain that $I_i \circ \mathrm{d}_h^{i-1} R_{i-1}\phi_{i-1} = \mathrm{d}^{i-1}\phi_{i-1} = I_i \boldsymbol{v}_i^h$. Therefore we can find $d_h^{i-1} R_{i-1}\phi_{i-1} = \boldsymbol{v}_i^h$. Thus we can get the conclusion.

In order to derive the error estimates in the subsequent analysis, particularly those presented in Section 4, we review the following stability and interpolation error estimates for the projection and reconstruction operators.

Lemma 3.1. *For some ϕ is smoothness enough, there exists a constant C¿0, independent of h, such*

that

$$\|\mathcal{R}_i\phi\|_{\boldsymbol{L}^\infty(\Omega)} \leq C\|\phi\|_{\boldsymbol{L}^\infty(\Omega)}, \tag{3.1}$$

$$\|\mathcal{R}_i\phi - \phi\|_{\boldsymbol{L}^2(\Omega)} \leq Ch\|\phi\|_{\boldsymbol{H}^1(\Omega)}, \tag{3.2}$$

$$\|\nabla \times \phi - \nabla_h \times \mathcal{R}_2\phi\|_{\boldsymbol{L}^2(\Omega)} \leq Ch\|\phi\|_{\boldsymbol{H}^2(\Omega)}, \tag{3.3}$$

$$\|\mathcal{I}_{RT}\phi - \phi\|_{\boldsymbol{L}^2(\Omega)} \leq Ch\|\phi\|_{\boldsymbol{H}^1(\Omega)}. \tag{3.4}$$

Proof. For the cases $i = 0, 1, 2$, (3.18) has been reported in [13, 14], as for the case $i = 3$, the interpolation error estimate can be easily get according to the lowest order DG-method. (3.19)-(3.20) have been reported in [40]. □

Next, we introduce the following algebraic identity reported in [25].

Lemma 3.2. *Let $\mathring{V}_h^0$,$\mathring{V}_h^1$ and $\mathring{V}_h^2$ denote the subspace of V_h^0, V_h^1 and V_h^2 apparently such that the functions in V_h^0, $\mathring{V}_h^1$ and $\mathring{V}_h^2$ are 0 on the degrees of freedom located on the boundary, then the following algebraic identities hold true*

$$\left\langle \nabla_h \times \boldsymbol{v}_1^h, \boldsymbol{w}_2^h \right\rangle_\Omega = \left\langle \nabla_h \times \boldsymbol{w}_2^h, \boldsymbol{v}_1^h \right\rangle_\Omega, \forall \boldsymbol{w}_2^h \in \mathring{V}_h^2 \tag{3.5}$$

$$\left\langle \nabla_h q_3^h, \boldsymbol{w}_2^h \right\rangle_\Omega = -\left\langle \nabla_h \cdot \boldsymbol{w}_2^h, q_3^h \right\rangle_\Omega, \ \forall q_3^h \in V_h^3, \ \boldsymbol{w}_2^h \in \mathring{V}_h^2 \tag{3.6}$$

$$\left\langle \nabla_h q_0^h, \boldsymbol{w}_1^h \right\rangle_\Omega = -\left\langle \nabla_h \cdot \boldsymbol{w}_1^h, q_0^h \right\rangle_\Omega, \forall q_0^h \in \mathring{V}_h^0 \tag{3.7}$$

We recall the following discrete Poincare inequality reported in [13].

Lemma 3.3. *(Discrete Poincare inequality) Let $i = 0, ..., 3$ and $\boldsymbol{v}_i^h \in V_h^i$, then there exists a constant $C > 0$, such that*

$$\inf_{\boldsymbol{z}_h \in \mathrm{Ker}(\mathrm{d}_h^i)} \|\boldsymbol{v}_i^h - \boldsymbol{z}_h\|_{\boldsymbol{L}^2(\Omega)} \leq C\|\mathrm{d}_h^i \boldsymbol{v}_i^h\|_{\boldsymbol{L}^2(\Omega)} \tag{3.8}$$

We can immediately get the following form of Discrete Poincare inequality.

Corollary 3.1. *Let $\boldsymbol{v}_1^h \in V_h^1$ such that $\left\langle \boldsymbol{v}_i^h, \nabla_h Q_0^h \right\rangle_\Omega$, $\forall Q_0^h \in V_h^0$, there exists a constant $C > 0$, such that the following inequality holds*

$$\|\boldsymbol{v}_1^h\|_{\boldsymbol{L}^2(\Omega)} \leq C\|\nabla_h \times \boldsymbol{v}_1^h\|_{\boldsymbol{L}^2(\Omega)} \tag{3.9}$$

Proof. By virtue of (3.8) and the fact that the discrete de Rahm complex is exact, there exists a $\boldsymbol{z}_h \in \mathrm{Ker}(\mathrm{d}_h^1)$ and $\phi_0^h \in V_h^0$ such that $\boldsymbol{z}_h = \nabla_h \phi_0^h$, we have

$$\begin{aligned}
\|\boldsymbol{v}_i^h - \boldsymbol{z}_h\|_{\boldsymbol{L}^2(\Omega)}^2 &= \|\boldsymbol{v}_1^h\|_{\boldsymbol{L}^2(\Omega)}^2 + \|\boldsymbol{z}_h\|_{\boldsymbol{L}^2(\Omega)}^2 - 2\left\langle \boldsymbol{v}_1^h, \boldsymbol{z}_h \right\rangle_\Omega \\
&= \|\boldsymbol{v}_1^h\|_{\boldsymbol{L}^2(\Omega)}^2 + \|\boldsymbol{z}_h\|_{\boldsymbol{L}^2(\Omega)}^2 - 2\left\langle \boldsymbol{v}_1^h, \nabla_h \phi_0^h \right\rangle_\Omega \\
&= \|\boldsymbol{v}_1^h\|_{\boldsymbol{L}^2(\Omega)}^2 + \|\boldsymbol{z}_h\|_{\boldsymbol{L}^2(\Omega)}^2 \\
&\leq C\|\nabla_h \times \boldsymbol{v}_1^h\|_{\boldsymbol{L}^2(\Omega)}^2.
\end{aligned} \tag{3.10}$$

Thus we can show that $\|\boldsymbol{v}_1^h\|_{\boldsymbol{L}^2(\Omega)} \leq C\|\nabla_h \times \boldsymbol{v}_1^h\|_{\boldsymbol{L}^2(\Omega)}$. □

Based on the discrete de Rham complex and operators established above, we now propose the pressure-robust semi-discrete dual-field MAC-Yee scheme. By utilizing the dual-field structure and the staggered grid, the scheme naturally enforces the discrete divergence-free constraint for both velocity fields. Furthermore, the spatial discretization of the nonlinear terms is carefully designed to be energy-preserving and helicity-preserving, maintaining the fundamental physics of the Navier-Stokes equations without introducing numerical dissipation.

$$\partial_t \boldsymbol{u}_1^h + \boldsymbol{\omega}_1^h \times \boldsymbol{u}_1^h + \frac{1}{Re}\nabla_h \times \boldsymbol{\omega}_2^h + \nabla_h P_0^h = \mathcal{R}_1 \boldsymbol{f}, \tag{3.11}$$

$$\nabla_h \times \boldsymbol{u}_1^h - \boldsymbol{\omega}_2^h = 0, \tag{3.12}$$

$$\nabla_h \cdot \boldsymbol{u}_1^h = 0, \tag{3.13}$$

$$\partial_t \boldsymbol{u}_2^h + \boldsymbol{\omega}_2^h \times \boldsymbol{u}_2^h + \frac{1}{Re}\nabla_h \times \boldsymbol{\omega}_1^h + \nabla_h P_3^h = \mathcal{I}_{RT} \boldsymbol{f}, \tag{3.14}$$

$$\nabla_h \times \boldsymbol{u}_2^h - \boldsymbol{\omega}_1^h = 0, \tag{3.15}$$

$$\nabla_h \cdot \boldsymbol{u}_2^h = 0. \tag{3.16}$$

The exterior force term $\mathcal{I}_{RT}\boldsymbol{f} \in V_h^2$ is defined as $(\mathcal{I}_{RT}\boldsymbol{f}, \boldsymbol{v}_2^h) = (\boldsymbol{f}, I_2 \boldsymbol{v}_2^h)$.

By applying the algebraic identities established in Lemma 3.2, we can derive the discrete weak formulation of the dual-field system (3.17)–(3.22). It is worth noting that the resulting discrete weak form maintains a similar structure to the finite element discretization based on the discrete de Rham complex as discussed in [44]. This structural consistency facilitates the application of analytical tools for proving the stability and error estimates of the scheme.

$$\left\langle \frac{\partial \boldsymbol{u}_1^h}{\partial t}, \boldsymbol{\epsilon}_1^h \right\rangle_\Omega + \left\langle \boldsymbol{\omega}_1^h \times \boldsymbol{u}_1^h, \boldsymbol{\epsilon}_1^h \right\rangle_\Omega + \frac{1}{Re}\left\langle \boldsymbol{\omega}_2^h, \nabla_h \times \boldsymbol{\epsilon}_1^h \right\rangle_\Omega + \left\langle \nabla_h P_0^h, \boldsymbol{\epsilon}_1^h \right\rangle_\Omega = \left\langle \mathcal{R}_1 \boldsymbol{f}, \boldsymbol{\epsilon}_1^h \right\rangle_\Omega \quad \forall \boldsymbol{\epsilon}_1^h \in V_h^1, \tag{3.17}$$

$$\left\langle \nabla_h \times \boldsymbol{u}_1^h, \boldsymbol{\epsilon}_2^h \right\rangle_\Omega - \left\langle \boldsymbol{\omega}_2^h, \boldsymbol{\epsilon}_2^h \right\rangle_\Omega = 0 \quad \forall \boldsymbol{\epsilon}_2^h \in V_h^2, \tag{3.18}$$

$$\left\langle \boldsymbol{u}_1^h, \nabla_h \epsilon_0^h \right\rangle_\Omega = 0 \quad \forall \epsilon_0^h \in V_h^0, \tag{3.19}$$

$$\left\langle \frac{\partial \boldsymbol{u}_2^h}{\partial t}, \boldsymbol{\epsilon}_2^h \right\rangle_\Omega + \left\langle \boldsymbol{\omega}_2^h \times \boldsymbol{u}_2^h, \boldsymbol{\epsilon}_2^h \right\rangle_\Omega + \frac{1}{Re}\left\langle \nabla_h \times \boldsymbol{\omega}_1^h, \boldsymbol{\epsilon}_2^h \right\rangle_\Omega - \left\langle P_3^h, \nabla_h \cdot \boldsymbol{\epsilon}_2^h \right\rangle_\Omega = \left\langle \mathcal{I}_{RT} \boldsymbol{f}, \boldsymbol{\epsilon}_2^h \right\rangle_\Omega \quad \forall \boldsymbol{\epsilon}_2^h \in V_h^2, \tag{3.20}$$

$$\left\langle \boldsymbol{u}_2^h, \nabla_h \times \boldsymbol{\epsilon}_1^h \right\rangle_\Omega - \left\langle \boldsymbol{\omega}_1^h, \boldsymbol{\epsilon}_1^h \right\rangle_\Omega = 0 \quad \forall \boldsymbol{\epsilon}_1^h \in V_h^1, \tag{3.21}$$

$$\left\langle \nabla_h \cdot \boldsymbol{u}_2^h, \epsilon_3^h \right\rangle_\Omega = 0 \quad \forall \epsilon_3^h \in V_h^3. \tag{3.22}$$

4 Pressure-robust semi-discrete error estimate

In this section, we present a rigorous error analysis for the proposed semi-discrete dual-field MAC-Yee scheme. The primary objective is to establish a pressure-robust error estimate, ensuring that the velocity error is independent of the pressure approximation and remains unaffected by potential high pressure gradients. This property is particularly critical for maintaining the accuracy and stability of numerical simulations in high-Reynolds-number regimes.

Firstly, we introduce the following priori estimates of the numerical solutions. Taking $\boldsymbol{\epsilon}_1^h = \boldsymbol{u}_1^h$ in (3.17), then we obtain the following energy identity

$$\frac{1}{2}\partial_t \|\boldsymbol{u}_1^h\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{1}{Re}\|\boldsymbol{\omega}_2^h\|^2_{\boldsymbol{L}^2(\Omega)} = \left\langle \mathcal{R}_1 \boldsymbol{f}, \boldsymbol{u}_1^h \right\rangle_\Omega \tag{4.1}$$

Taking $\boldsymbol{\epsilon}_2^h = \boldsymbol{u}_2^h$ in (3.20), it follows

$$\frac{1}{2}\partial_t \|\boldsymbol{u}_2^h\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{1}{Re}\|\boldsymbol{\omega}_1^h\|^2_{\boldsymbol{L}^2(\Omega)} = \left\langle \mathcal{I}_{RT} \boldsymbol{f}, \boldsymbol{u}_2^h \right\rangle_\Omega . \tag{4.2}$$

Calculating over $[0, T]$, by virtue of discrete Poincare inequality, we have the following priori estimates

$$\|\boldsymbol{u}_1^h\|_{L^\infty(0,T;\boldsymbol{L}^2(\Omega))} \le C, \quad \|\boldsymbol{\omega}_2^h\|_{L^2(0,T;\boldsymbol{L}^2(\Omega))} \le C,$$
$$\|\boldsymbol{u}_2^h\|_{L^\infty(0,T;\boldsymbol{L}^2(\Omega))} \le C, \quad \|\boldsymbol{\omega}_1^h\|_{L^2(0,T;\boldsymbol{L}^2(\Omega))} \le C.$$

Approximation operators

To establish the error estimates between the continuous solutions $(\boldsymbol{u}, \boldsymbol{\omega}, P)$ and the discrete solutions $(\boldsymbol{u}_i^h, \boldsymbol{\omega}_h^i, P_h^i)$, it is essential to construct suitable projection operators from the continuous spaces to the discrete ones. While the canonical interpolation operators $\{\mathcal{R}_i\}_{i=0}^3$ are naturally defined via the de Rham complex, directly employing them in the error analysis of the momentum equation would introduce several uncontrollable terms. Specifically, the coupling between the velocity error and the pressure gradient would fail to vanish. To circumvent this, we introduce a specific set of operators $\{\Pi^i\}_{i=0}^3$. In particular, Π^1 is designed based on a discrete Hodge decomposition to ensure that the irrotational part of the velocity field does not interfere with the momentum balance, thereby yielding a robust error estimate [14]. The operators Π^i is defined as:

- Π^0:$\mathcal{R}_0$.
- Π^1:For some $\boldsymbol{v} \in V^1$, according to [14], there exists a discrete Hodge decomposition

  $$\mathcal{R}_1 \boldsymbol{v} = \boldsymbol{w}_h + \boldsymbol{w}_h^\perp,$$

  such that $\boldsymbol{w}_h \in \mathrm{Im}(\mathrm{d}_h^0)$ and $\boldsymbol{w}_h^\perp \in (\mathrm{Im}(\mathrm{d}_h^0))^\perp$, then we define that $\Pi^1 \boldsymbol{v} = \boldsymbol{w}_h^\perp$. Since the discrete complex is exact, $\nabla_h \times \boldsymbol{w}^h = 0$, thus it follows $\nabla_h \times \Pi^1 \boldsymbol{v} = \nabla_h \times \mathcal{R}_1 \boldsymbol{v}$. Moreover, if $\boldsymbol{v} \in \mathrm{Im}(\mathrm{d}^0)$, according to [14], then there exists an error estimate $\|\boldsymbol{v} - \Pi^1 \boldsymbol{v}\|_{\boldsymbol{L}^2(\Omega)} \le Ch\|\boldsymbol{v}\|_{\boldsymbol{H}^1(\Omega)}$.
- Π^2: $\mathcal{R}_2$.
- Π^3: L^2 projection from V^3 to V_h^3, i.e. $\mathcal{R}_3$.

Error estimate

Theorem 4.1. *(Pressure-robust error estimate) Let$(\boldsymbol{u}, \boldsymbol{\omega}, P)$ be the sufficiently smooth solution to the continuous dual-field Navier-Stokes equations, $(\boldsymbol{u}_h^1, \boldsymbol{u}_h^2, \boldsymbol{\omega}_h^1, \boldsymbol{\omega}_h^2, P_h^0, P_h^3)$ be the solutions of the semi-discrete dual-field MAC-Yee scheme. Suppose that the assumptions of theorem 2.1 hold true, there*

exists a constant $C > 0$, independent of the mesh size h, the pressure P, such that for any $T > 0$

$$\begin{aligned}&\frac{1}{2}(\|e^1_{\boldsymbol{u}}(T)\|^2_{\boldsymbol{L}^2(\Omega)} + \|e^2_{\boldsymbol{u}}(T)\|^2_{\boldsymbol{L}^2(\Omega)}) + \frac{5}{8Re}\|e^1_{\boldsymbol{\omega}}\|^2_{L^2(0,T;\boldsymbol{L}^2(\Omega))} + \frac{3}{4Re}\|e^2_{\boldsymbol{\omega}}\|^2_{L^2(0,T;\boldsymbol{L}^2(\Omega))}\\ &\quad \le Ch^2 exp(T)\int_0^T \mathbb{S}(\boldsymbol{u}_1(s), \boldsymbol{u}_2(s), \boldsymbol{\omega}_1(s), \boldsymbol{\omega}_2(s))\,\mathrm{d}s,\end{aligned} \tag{4.3}$$

where

$$\begin{aligned}\mathbb{S}(\boldsymbol{u},\boldsymbol{\omega}) := &\|\partial_t \boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{\omega}\times\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{\omega}\|^2_{\boldsymbol{H}^1(\Omega)}\|\boldsymbol{u}\|^2_{\boldsymbol{L}^\infty(\Omega)} + \|\boldsymbol{\omega}\|^2_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)}\\ &+ \frac{1}{Re}\|\nabla\times\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} + \frac{1}{Re}\|\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)}.\end{aligned}$$

and

$$\begin{aligned}&e^1_{\boldsymbol{u}} = \boldsymbol{u}^h_1 - \Pi^1\boldsymbol{u}, \quad e^2_{\boldsymbol{\omega}} = \boldsymbol{\omega}^h_2 - \Pi^2\boldsymbol{\omega}, \quad e^0_P = P^h_0 - \Pi^0 P,\\ &e^2_{\boldsymbol{u}} = \boldsymbol{u}^h_2 - \Pi^2\boldsymbol{u}, \quad e^1_{\boldsymbol{\omega}} = \boldsymbol{\omega}^h_1 - \Pi^1\boldsymbol{\omega}, \quad e^3_P = P^h_3 - \Pi^3 P.\end{aligned}$$

Proof. Firstly, we consider (3.17)-(3.19) and (3.17)-(3.19) can be written as

$$\begin{aligned}&\left\langle \frac{\partial e^1_{\boldsymbol{u}}}{\partial t}, \boldsymbol{\epsilon}^h_1\right\rangle_\Omega + \frac{1}{Re}\left\langle \nabla_h\times e^1_{\boldsymbol{u}}, \nabla_h\times\boldsymbol{\epsilon}^h_1\right\rangle_\Omega + \left\langle \nabla_h P^h_0, \boldsymbol{\epsilon}^h_1\right\rangle_\Omega\\ &= \left\langle \mathcal{R}_1\boldsymbol{f}, \boldsymbol{\epsilon}^h_1\right\rangle_\Omega - \left\langle \boldsymbol{\omega}^h_1\times\boldsymbol{u}^h_1, \boldsymbol{\epsilon}^h_1\right\rangle_\Omega - \left\langle \frac{\partial \Pi^1\boldsymbol{u}}{\partial t}, \boldsymbol{\epsilon}^h_1\right\rangle_\Omega - \frac{1}{Re}\left\langle \nabla_h\times\Pi^1\boldsymbol{u}, \nabla_h\times\boldsymbol{\epsilon}^h_1\right\rangle_\Omega.\end{aligned} \tag{4.4}$$

Let $\boldsymbol{\epsilon}^h_1 = e^1_{\boldsymbol{u}}$, using the definition of Π^1 and (3.19), then

$$\begin{aligned}\left\langle \nabla_h P^h_0, e^1_{\boldsymbol{u}}\right\rangle_\Omega &= \left\langle \nabla_h P^h_0, \boldsymbol{u}^h_1\right\rangle_\Omega - \left\langle \nabla_h P^h_0, \Pi^1\boldsymbol{u}\right\rangle_\Omega\\ &= 0.\end{aligned} \tag{4.5}$$

Thus (4.4) can be written as

$$\frac{1}{2}\frac{\partial}{\partial_t}\left\langle e^1_{\boldsymbol{u}}, e^1_{\boldsymbol{u}}\right\rangle_\Omega + \frac{1}{Re}\|\nabla_h\times e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} = \mathcal{A}_{1,1} + \mathcal{A}_{1,2} + \mathcal{A}_{1,3} + \mathcal{A}_{1,4}. \tag{4.6}$$

The remainder terms are

$$\begin{aligned}\mathcal{A}_{1,1} &= \left\langle \partial_t(\mathcal{R}_1\boldsymbol{u} - \Pi^1\boldsymbol{u}), e^1_{\boldsymbol{u}}\right\rangle_\Omega,\\ \mathcal{A}_{1,2} &= \left\langle \mathcal{R}_1(\boldsymbol{\omega}_1\times\boldsymbol{u}), e^1_{\boldsymbol{u}}\right\rangle_\Omega - \left\langle \boldsymbol{\omega}^h_1\times\Pi^1\boldsymbol{u}, e^1_{\boldsymbol{u}}\right\rangle_\Omega,\\ \mathcal{A}_{1,3} &= \frac{1}{Re}\left\langle \mathcal{R}_1(\nabla\times\nabla\times\boldsymbol{u}), e^1_{\boldsymbol{u}}\right\rangle_\Omega - \frac{1}{Re}\left\langle \nabla_h\times\Pi^1\boldsymbol{u}, \nabla_h\times e^1_{\boldsymbol{u}}\right\rangle_\Omega,\\ \mathcal{A}_{1,4} &= \left\langle \mathcal{R}_1\nabla P, e^1_{\boldsymbol{u}}\right\rangle_\Omega.\end{aligned}$$

(3.20) can be rewritten as

$$\begin{aligned}&\left\langle \frac{\partial e^2_{\boldsymbol{u}}}{\partial t}, \boldsymbol{\epsilon}^h_2\right\rangle_\Omega + \frac{1}{Re}\left\langle \nabla_h\times e^1_{\boldsymbol{\omega}}, \boldsymbol{\epsilon}^h_2\right\rangle_\Omega - \left\langle P^h, \nabla_h\cdot\boldsymbol{\epsilon}^h_2\right\rangle_\Omega\\ &= \left\langle \frac{\partial(\mathcal{I}_{RT}\boldsymbol{u} - \Pi^2\boldsymbol{u})}{\partial t}, \boldsymbol{\epsilon}^h_2\right\rangle_\Omega\\ &\quad + \left\langle \mathcal{I}_{RT}(\boldsymbol{\omega}\times\boldsymbol{u}), \boldsymbol{\epsilon}^h_2\right\rangle_\Omega - \left\langle \boldsymbol{\omega}^h_2\times\boldsymbol{u}^h_2, \boldsymbol{\epsilon}^h_2\right\rangle_\Omega\\ &\quad + \frac{1}{Re}\left\langle \mathcal{I}_{RT}(\nabla\times\omega_1) - \nabla_h\times\Pi^1\omega_1, \boldsymbol{\epsilon}^h_2\right\rangle_\Omega\\ &\quad + \left\langle \mathcal{I}_{RT}P, \nabla_h\cdot\boldsymbol{\epsilon}^h_2\right\rangle_\Omega.\end{aligned} \tag{4.7}$$

Let $\boldsymbol{\epsilon}_2^h = e_{\boldsymbol{u}}^2$ in (4.7), we obtain that

$$\begin{aligned}
&\frac{1}{2}\frac{\partial}{\partial_t}\left\langle e_{\boldsymbol{u}}^2, e_{\boldsymbol{u}}^2\right\rangle_\Omega + \frac{1}{Re}\left\langle \nabla_h \times e_{\boldsymbol{\omega}}^1, e_{\boldsymbol{u}}^2\right\rangle_\Omega \\
&= \left\langle \frac{\partial(\mathcal{I}_{RT}\boldsymbol{u} - \Pi^2\boldsymbol{u})}{\partial t}, e_{\boldsymbol{u}}^2\right\rangle_\Omega \\
&\quad + \left\langle \mathcal{I}_{RT}(\boldsymbol{\omega}\times\boldsymbol{u}), e_{\boldsymbol{u}}^2\right\rangle_\Omega - \left\langle \boldsymbol{\omega}_2^h \times \boldsymbol{u}_2^h, e_{\boldsymbol{u}}^2\right\rangle_\Omega \\
&\quad + \frac{1}{Re}\left\langle \mathcal{I}_{RT}(\nabla\times\omega_1) - \nabla_h \times \Pi^1\omega_1, e_{\boldsymbol{u}}^2\right\rangle_\Omega \\
&\quad + \left\langle \mathcal{I}_{RT}P, \nabla_h \cdot e_{\boldsymbol{u}}^2\right\rangle_\Omega
\end{aligned} \tag{4.8}$$

The second term of the left hand side of (4.8) can be written as

$$\begin{aligned}
\frac{1}{Re}\left\langle \nabla_h \times e_{\boldsymbol{\omega}}^1, e_{\boldsymbol{u}}^2\right\rangle_\Omega &= \frac{1}{Re}\left\langle \nabla_h \times e_{\boldsymbol{\omega}}^1, \boldsymbol{u}_2^h\right\rangle_\Omega - \frac{1}{Re}\left\langle \nabla_h \times e_{\boldsymbol{\omega}}^1, \Pi^2\boldsymbol{u}\right\rangle_\Omega \\
&= \frac{1}{Re}\left\langle \boldsymbol{\omega}_1^h, e_{\boldsymbol{\omega}}^1\right\rangle_\Omega - \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, \nabla_h \times \Pi^2\boldsymbol{u}\right\rangle_\Omega \\
&= \frac{1}{Re}\left\langle \boldsymbol{\omega}_1^h, e_{\boldsymbol{\omega}}^1\right\rangle_\Omega - \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, \Pi^1\boldsymbol{\omega}\right\rangle_\Omega \\
&\quad + \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, \nabla\times\boldsymbol{u} - \nabla_h \times \Pi^2\boldsymbol{u}\right\rangle_\Omega \\
&\quad + \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, \Pi^1\boldsymbol{\omega} - \nabla\times\boldsymbol{u}\right\rangle_\Omega \\
&= \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, e_{\boldsymbol{\omega}}^1\right\rangle_\Omega \\
&\quad + \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, \nabla\times\boldsymbol{u} - \nabla_h \times \Pi^2\boldsymbol{u}\right\rangle_\Omega \\
&\quad + \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, \Pi^1\boldsymbol{\omega} - \nabla\times\boldsymbol{u}\right\rangle_\Omega .
\end{aligned} \tag{4.9}$$

Therefore, from (4.8) we have

$$\begin{aligned}
&\frac{1}{2}\frac{\partial}{\partial_t}\left\langle e_{\boldsymbol{u}}^2, e_{\boldsymbol{u}}^2\right\rangle_\Omega + \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, e_{\boldsymbol{\omega}}^1\right\rangle_\Omega \\
&= \left\langle \frac{\partial(\mathcal{I}_{RT}\boldsymbol{u} - \Pi^2\boldsymbol{u})}{\partial t}, e_{\boldsymbol{u}}^2\right\rangle_\Omega \\
&\quad + \left\langle \mathcal{I}_{RT}(\boldsymbol{\omega}\times\boldsymbol{u}), e_{\boldsymbol{u}}^2\right\rangle_\Omega - \left\langle \boldsymbol{\omega}_2^h \times \boldsymbol{u}_2^h, e_{\boldsymbol{u}}^2\right\rangle_\Omega \\
&\quad + \frac{1}{Re}\left\langle \mathcal{I}_{RT}(\nabla\times\omega_1) - \nabla_h \times \Pi^1\omega_1, e_{\boldsymbol{u}}^2\right\rangle_\Omega \\
&\quad + \left\langle \mathcal{I}_{RT}P, \nabla_h \cdot e_{\boldsymbol{u}}^2\right\rangle_\Omega \\
&\quad + \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, \nabla_h \times \Pi^2\boldsymbol{u} - \nabla\times\boldsymbol{u}\right\rangle_\Omega \\
&\quad + \frac{1}{Re}\left\langle e_{\boldsymbol{\omega}}^1, \nabla\times\boldsymbol{u} - \Pi^1\boldsymbol{\omega}\right\rangle_\Omega =: \sum_{j=1}^{6}\mathcal{A}_{2,j}
\end{aligned} \tag{4.10}$$

Add (4.6) and (4.10), we obtain the total error equation of the discrete dual-field system

$$\frac{1}{2}\partial_t(\|e_{\boldsymbol{u}}^1\|_{\boldsymbol{L}^2(\Omega)}^2 + \|e_{\boldsymbol{u}}^2\|_{\boldsymbol{L}^2(\Omega)}^2) + \frac{1}{Re}(\|e_{\boldsymbol{\omega}}^2\|_{\boldsymbol{L}^2(\Omega)}^2 + \|e_{\boldsymbol{\omega}}^1\|_{\boldsymbol{L}^2(\Omega)}^2) = \sum_{j=1}^{4}\mathcal{A}_{1,j} + \sum_{j=1}^{6}\mathcal{A}_{2,j}. \tag{4.11}$$

Then we will estimate the right terms of (4.11).

- $\mathcal{A}_{1,1}$: According to the Cauchy-Schwarz inequality, Young inequality and interpolation error estimates，we have

$$\begin{aligned}\mathcal{A}_{1,1} &\leq \|\partial_t(\boldsymbol{u}-\Pi^1\boldsymbol{u})\|_{\boldsymbol{L}^2(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} + \|\partial_t(\boldsymbol{u}-\mathcal{R}_1\boldsymbol{u})\|_{\boldsymbol{L}^2(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\leq Ch^2\|\partial_t\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)}.\end{aligned} \tag{4.12}$$

- $\mathcal{A}_{1,2}$: By applying the Hölder inequalityinequality，Young inequality and interpolation error estimates, it follows

$$\begin{aligned}\mathcal{A}_{1,2} &= \left\langle \mathcal{R}_1(\boldsymbol{\omega}\times\boldsymbol{u}) - \boldsymbol{\omega}\times\boldsymbol{u}, e^1_{\boldsymbol{u}}\right\rangle_\Omega + \left\langle (\boldsymbol{\omega}-\Pi^1\boldsymbol{\omega})\times\boldsymbol{u}, e^1_{\boldsymbol{u}}\right\rangle_\Omega \\ &\quad + \left\langle \Pi^1\boldsymbol{\omega}\times(\boldsymbol{u}-\Pi^1\boldsymbol{u}), e^1_{\boldsymbol{u}}\right\rangle_\Omega + \left\langle e^1_{\boldsymbol{\omega}}\times\Pi^1\boldsymbol{u}, e^1_{\boldsymbol{u}}\right\rangle_\Omega + \left\langle e^1_{\boldsymbol{u}}\times\boldsymbol{\omega}^h_1, e^1_{\boldsymbol{u}}\right\rangle_\Omega \\ &\leq \|\mathcal{R}_1(\boldsymbol{\omega}\times\boldsymbol{u}) - \boldsymbol{\omega}\times\boldsymbol{u}\|_{\boldsymbol{L}^2(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\quad + \|\boldsymbol{u}\|_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{\omega}-\Pi^1\boldsymbol{\omega}\|_{\boldsymbol{L}^2(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} + \|\Pi^1\boldsymbol{\omega}\|_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{u}-\Pi^1\boldsymbol{u}\|_{\boldsymbol{L}^2(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\quad + \|\Pi^1\boldsymbol{u}\|_{\boldsymbol{L}^\infty(\Omega)}\|e^1_{\boldsymbol{\omega}}\|_{\boldsymbol{L}^2(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\leq Ch\|\boldsymbol{\omega}\times\boldsymbol{u}\|_{\boldsymbol{H}^1(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} + Ch\|\boldsymbol{u}\|_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{\omega}\|_{\boldsymbol{H}^1(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\quad + Ch\|\boldsymbol{\omega}\|_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{u}\|_{\boldsymbol{H}^1(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} + \|\boldsymbol{u}\|_{\boldsymbol{L}^\infty(\Omega)}\|e^1_{\boldsymbol{\omega}}\|_{\boldsymbol{L}^2(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\leq C(\|\boldsymbol{\omega}\times\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{u}\|^2_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{\omega}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)}\|\boldsymbol{\omega}\|^2_{\boldsymbol{L}^\infty(\Omega)})h^2 \\ &\quad + (C\|\boldsymbol{u}\|^2_{\boldsymbol{L}^\infty(\Omega)} + 3)\|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{1}{8Re}\|e^1_{\boldsymbol{\omega}}\|^2_{\boldsymbol{L}^2(\Omega)}\end{aligned} \tag{4.13}$$

- $\mathcal{A}_{1,3}$: By applying the Cauchy-Schwartz inequality, Young inequality and [13, (6.12), Theorem 10], the remainder term $\mathcal{A}_{1,3}$ can be controlled as

$$\begin{aligned}\mathcal{A}_{1,3} &= \frac{1}{Re}\left\langle \mathcal{R}_1(\nabla\times\nabla\boldsymbol{u}), e^1_{\boldsymbol{u}}\right\rangle_\Omega - \frac{1}{Re}\left\langle \nabla\times\nabla\times\boldsymbol{u}, I_1 e^1_{\boldsymbol{u}}\right\rangle_\Omega \\ &\quad + \frac{1}{Re}\left\langle \nabla\times\nabla\times\boldsymbol{u}, I_1 e^1_{\boldsymbol{u}}\right\rangle_\Omega - \frac{1}{Re}\left\langle \mathcal{R}_2\nabla\times\boldsymbol{u}, \nabla_h\times e^1_{\boldsymbol{u}}\right\rangle_\Omega \\ &\leq Ch\frac{1}{Re}\|\nabla\times\nabla\times\boldsymbol{u}\|_{\boldsymbol{H}^2(\Omega)}\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\quad + Ch\frac{1}{Re}(\|\nabla\times\boldsymbol{u}\|_{\boldsymbol{H}^1(\Omega)} + \|\nabla\times\boldsymbol{u}\|_{\boldsymbol{H}^2(\Omega)})(\|e^1_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} + \|e^1_{\boldsymbol{u}}\|_{V^1_h}) \\ &\leq Ch^2\frac{1}{Re}\|\nabla\times\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} + \frac{1}{Re}\|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} \\ &\quad + Ch^2\frac{1}{Re}\|\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} + \frac{1}{Re}\|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} \\ &\quad + \frac{1}{8Re}(\|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} + \|\nabla_h\times(\boldsymbol{u}-\Pi^1\boldsymbol{u})\|^2_{\boldsymbol{L}^2(\Omega)}) \\ &= Ch^2\frac{1}{Re}\|\nabla\times\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} + Ch^2\frac{1}{Re}\|\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} \\ &\quad + \frac{9}{8Re}\|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{1}{8Re}\|e^2_{\boldsymbol{\omega}}\|^2_{\boldsymbol{L}^2(\Omega)}.\end{aligned} \tag{4.14}$$

- $\mathcal{A}_{1,4}$: According to the commutative property of the continuous-discrete de Rahm diagram , (3.19) and the definition of Π^1, we have

$$\mathcal{A}_{1,4} = \left\langle \nabla_h I_0 P, e^1_{\boldsymbol{u}}\right\rangle_\Omega = 0. \tag{4.15}$$

- $\mathcal{A}_{2,1}$: Similar to (4.12), we have

$$\mathcal{A}_{2,1} \leq Ch^2\|\partial_t \boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|e^2_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)}. \tag{4.16}$$

- $\mathcal{A}_{2,2}$: Similar to (4.13)， it follows

$$\begin{aligned}\mathcal{A}_{2,2} \leq & C(\|\boldsymbol{\omega}\times\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{u}\|^2_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{\omega}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)}\|\boldsymbol{\omega}\|^2_{\boldsymbol{L}^\infty(\Omega)})h^2 \\ & (C\|\boldsymbol{u}\|^2_{\boldsymbol{L}^\infty(\Omega)} + 3)\|e^2_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{1}{8Re}\|e^2_{\boldsymbol{\omega}}\|^2_{\boldsymbol{L}^2(\Omega)}\end{aligned} \tag{4.17}$$

- $\mathcal{A}_{2,3}$: By applying the interpolation error estimates, Young inequality and [13, (6.12), Theorem 10], we have

$$\begin{aligned}\mathcal{A}_{2,3} &= \frac{1}{Re}\left\langle \mathcal{I}_{RT}(\nabla\times\boldsymbol{\omega}) - \nabla\times\boldsymbol{\omega}, e^2_{\boldsymbol{u}}\right\rangle_\Omega + \frac{1}{Re}\left\langle \nabla\times\boldsymbol{\omega} - \nabla_h\times\Pi^1\boldsymbol{\omega}, e^2_{\boldsymbol{u}}\right\rangle_\Omega \\ &\leq Ch\frac{1}{Re}\|\nabla\times\boldsymbol{\omega}\|_{\boldsymbol{H}^1(\Omega)}\|e^2_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} + Ch\frac{1}{Re}\|\nabla\times\boldsymbol{\omega}\|_{\boldsymbol{H}^1(\Omega)}\|e^2_{\boldsymbol{u}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\leq C\frac{1}{Re}h^2\|\nabla\times\boldsymbol{\omega}\|^2_{\boldsymbol{H}^1(\Omega)} + \frac{1}{Re}\|e^2_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)}.\end{aligned} \tag{4.18}$$

- $\mathcal{A}_{2,4}$: Since Π^2 is a Fortin operator, we have

$$\mathcal{A}_{2,4} = 0. \tag{4.19}$$

- $\mathcal{A}_{2,5}$: Using the interpolation error estimates and Young inequality, it holds that

$$\begin{aligned}\mathcal{A}_{2,5} &\leq \frac{1}{Re}\|e^1_{\boldsymbol{\omega}}\|_{\boldsymbol{L}^2(\Omega)}\|\nabla_h\times\Pi^2\boldsymbol{u} - \nabla\times\boldsymbol{u}\|_{\boldsymbol{L}^2(\Omega)} \\ &\leq Ch\frac{1}{Re}\|e^1_{\boldsymbol{\omega}}\|_{\boldsymbol{L}^2(\Omega)}\|\boldsymbol{u}\|_{\boldsymbol{H}^2(\Omega)} \\ &\leq Ch^2\frac{1}{Re}\|\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} + \frac{1}{8Re}\|e^1_{\boldsymbol{\omega}}\|^2_{\boldsymbol{L}^2(\Omega)}\end{aligned} \tag{4.20}$$

- $\mathcal{A}_{2,6}$: Similar to (4.20), we obtain that

$$\begin{aligned}\mathcal{A}_{2,6} &= \frac{1}{Re}\left\langle e^1_{\boldsymbol{\omega}}, \boldsymbol{\omega} - \Pi^1\boldsymbol{\omega}\right\rangle_\Omega \\ &\leq Ch\frac{1}{Re}\|\boldsymbol{\omega}\|_{\boldsymbol{H}^2(\Omega)}\|e^1_{\boldsymbol{\omega}}\|_{\boldsymbol{L}^2(\Omega)} \\ &\leq Ch^2\frac{1}{Re}\|\boldsymbol{\omega}\|^2_{\boldsymbol{H}^2(\Omega)} + \frac{1}{8Re}\|e^1_{\boldsymbol{\omega}}\|^2_{\boldsymbol{L}^2(\Omega)}.\end{aligned} \tag{4.21}$$

Thanks to the estimate of the remainder terms, then there exists a constant $C > 0$, independent of $h, \Delta t$, P, P and Re, such that

$$\begin{aligned}&\frac{1}{2}\partial_t(\|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} + \|e^2_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)}) + \frac{5}{8Re}\|e^1_{\boldsymbol{\omega}}\|^2_{\boldsymbol{L}^2(\Omega)} + \frac{3}{4Re}\|e^2_{\boldsymbol{\omega}}\|^2_{\boldsymbol{L}^2(\Omega)}) \\ &\leq (4 + 2Re + \frac{9}{8Re})\|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} + (4 + 2Re + \frac{1}{Re})\|e^2_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} \\ &\quad + Ch^2(\|\partial_t\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{\omega}\times\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{\omega}\|^2_{\boldsymbol{H}^1(\Omega)}\|\boldsymbol{u}\|^2_{\boldsymbol{L}^\infty(\Omega)} + \|\boldsymbol{\omega}\|^2_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} \\ &\quad + \frac{1}{Re}\|\nabla\times\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} + \frac{1}{Re}\|\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} + \|\partial_t\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} \\ &\quad + \|\boldsymbol{\omega}\times\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{\omega}\|^2_{\boldsymbol{H}^1(\Omega)}\|\boldsymbol{u}\|^2_{\boldsymbol{L}^\infty(\Omega)} + \|\boldsymbol{\omega}\|^2_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \frac{1}{Re}\|\nabla\times\boldsymbol{\omega}\|^2_{\boldsymbol{H}^1(\Omega)}) \\ &\leq C(\|e^1_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)} + \|e^2_{\boldsymbol{u}}\|^2_{\boldsymbol{L}^2(\Omega)}) + Ch^2\mathbb{S}(\boldsymbol{u},\boldsymbol{\omega}).\end{aligned} \tag{4.22}$$

The term $\mathbb{S}(\boldsymbol{u},\boldsymbol{u},\boldsymbol{\omega},\boldsymbol{\omega})$ is defined as

$$\begin{aligned}\mathbb{S}(\boldsymbol{u},\boldsymbol{\omega}) := &\|\partial_t \boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{\omega}\times\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)} + \|\boldsymbol{\omega}\|^2_{\boldsymbol{H}^1(\Omega)}\|\boldsymbol{u}\|^2_{\boldsymbol{L}^\infty(\Omega)} + \|\boldsymbol{\omega}\|^2_{\boldsymbol{L}^\infty(\Omega)}\|\boldsymbol{u}\|^2_{\boldsymbol{H}^1(\Omega)}\\ &+ \frac{1}{Re}\|\nabla\times\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)} + \frac{1}{Re}\|\nabla\times\boldsymbol{u}\|^2_{\boldsymbol{H}^2(\Omega)}.\end{aligned}$$

Applying the Gronwall inequality, we have that

$$\begin{aligned}&\frac{1}{2}(\|e^1_{\boldsymbol{u}}(T)\|^2_{\boldsymbol{L}^2(\Omega)} + \|e^2_{\boldsymbol{u}}(T)\|^2_{\boldsymbol{L}^2(\Omega)}) + \frac{5}{8Re}\|e^1_{\boldsymbol{\omega}}\|^2_{L^2(0,T;\boldsymbol{L}^2(\Omega))} + \frac{3}{4Re}\|e^2_{\boldsymbol{\omega}}\|^2_{L^2(0,T;\boldsymbol{L}^2(\Omega))}\\ &\quad\le Ch^2\exp(CT)\int_0^T \mathbb{S}(\boldsymbol{u}(s),\boldsymbol{\omega}(s))\,\mathrm{d}s.\end{aligned} \tag{4.23}$$

□

5 Fully-discrete scheme and conservation of physical quantity

In this section, we construct the fully discrete MAC-Yee scheme by integrating the spatial discretization described in Section 3 with a staggered temporal integrator. The primary advantage of this formulation is its ability to advance the two velocity and vorticity fields on interleaved time grids, which leads to linear and decoupled algebraic systems at each time step .

A fundamental aspect of this chapter is the rigorous verification of the scheme's structure-preserving properties. We aim to demonstrate that the proposed method exactly mimics the key conservation laws of the continuum model at the discrete level. Specifically, we provide detailed proofs showing that the scheme conserves both discrete kinetic energy and helicity in the inviscid, unforced limit. Furthermore, we analyze the discrete dissipation rates in the viscous case to confirm that the scheme correctly captures the physical energy cascade.

5.1 Fully-discrete scheme

The fully-discrete scheme is defined as follows:

Integer time steps

Given $(\boldsymbol{\omega}_1^{h,k-1},\boldsymbol{u}_2^{h,k-1},\boldsymbol{f}^{k-\frac{1}{2}},\boldsymbol{\omega}_2^{h,k-\frac{1}{2}})\in V_h^1\times V_h^2\times\times\boldsymbol{L}^2(\Omega)\times V_h^2$, solve $(\boldsymbol{\omega}_1^{h,k},\boldsymbol{u}_2^{h,k},P_3^{h,k-\frac{1}{2}})\in V_h^1\times V_h^2\times V_h^3$ such that

$$\frac{\boldsymbol{u}_2^{h,k}-\boldsymbol{u}_2^{h,k-1}}{\Delta t} + \boldsymbol{\omega}_2^{h,k-\frac{1}{2}}\times\frac{\boldsymbol{u}_2^{h,k}+\boldsymbol{u}_2^{h,k-1}}{2} + \frac{1}{Re}\nabla_h\times\frac{\boldsymbol{\omega}_1^{h,k}+\boldsymbol{\omega}_1^{h,k-1}}{2} + \nabla_h P_3^{h,k-\frac{1}{2}} = \mathcal{I}_{RT}\boldsymbol{f}^{k-\frac{1}{2}}, \tag{5.1}$$

$$\nabla_h\times\boldsymbol{u}_2^{h,k} - \boldsymbol{\omega}_1^{h,k} = 0, \tag{5.2}$$

$$\nabla\cdot\boldsymbol{u}_2^{h,k} = 0. \tag{5.3}$$

Half-integer time steps

Given $(\boldsymbol{u}_1^{h,k-\frac{1}{2}},\boldsymbol{\omega}_2^{h,k-\frac{1}{2}},\boldsymbol{f}^k,\boldsymbol{\omega}_1^{h,k})\in V_h^1\times V_h^2\times\boldsymbol{L}^2(\Omega)\times V_h^1$, solve $(P_0^{h,k},\boldsymbol{u}_1^{h,k+\frac{1}{2}},\boldsymbol{\omega}_2^{h,k+\frac{1}{2}})\in V_h^0\times V_h^1\times V_h^2$ such that

$$\frac{\boldsymbol{u}_1^{h,k+\frac{1}{2}}-\boldsymbol{u}_1^{h,k-\frac{1}{2}}}{\Delta t} + \boldsymbol{\omega}_1^{h,k}\times\frac{\boldsymbol{u}_1^{h,k+\frac{1}{2}}+\boldsymbol{u}_1^{h,k-\frac{1}{2}}}{2} + \frac{1}{Re}\nabla_h\times\frac{\boldsymbol{\omega}_2^{h,k+\frac{1}{2}}+\boldsymbol{\omega}_2^{h,k-\frac{1}{2}}}{2} + \nabla_h P_0^{h,k} = \mathcal{R}_1\boldsymbol{f}^k, \tag{5.4}$$

$$\nabla_h \times \boldsymbol{u}_1^{h,k+\frac{1}{2}} - \boldsymbol{\omega}_2^{h,k+\frac{1}{2}} = 0, \tag{5.5}$$

$$\nabla_h \cdot \boldsymbol{u}_1^{h,k+\frac{1}{2}} = 0. \tag{5.6}$$

Starting procedure

To start the numerical algorithm, given the initial values $\boldsymbol{u}_2^{h,0}$ and $\boldsymbol{\omega}_1^{h,0}$ at integer time steps, we need to obtain the initial value $\boldsymbol{\omega}_2^{h,\frac{1}{2}}$ at the half-integer time step. We employ an explicit Euler scheme of the following form to compute $\boldsymbol{u}_1^{h,\frac{1}{2}}$ and $\boldsymbol{\omega}_2^{h,\frac{1}{2}}$.

Given $(\boldsymbol{u}_1^{h,0}, \boldsymbol{\omega}_1^{h,0}, \boldsymbol{\omega}_2^{h,0})$, solve $(\boldsymbol{u}_1^{h,\frac{1}{2}}, P_0^{h,0}, \boldsymbol{\omega}_2^{h,\frac{1}{2}})$ such that

$$\frac{\boldsymbol{u}_1^{h,\frac{1}{2}}}{\Delta t} + \nabla_h P_0^{h,0} = \mathcal{R}_1 \boldsymbol{f}^0 - \boldsymbol{\omega}_1^{h,0} \times \boldsymbol{u}_1^{h,0} - \frac{1}{Re}\nabla_h \times \boldsymbol{\omega}_2^{h,0} + \frac{\boldsymbol{u}_1^{h,0}}{\Delta t},$$
$$\nabla_h \cdot \boldsymbol{u}_1^{h,\frac{1}{2}} = 0,$$

and

$$\boldsymbol{\omega}_2^{h,\frac{1}{2}} = \nabla_h \times \boldsymbol{u}_1^{h,\frac{1}{2}}.$$

5.2 Kinetic energy conservation

We first demonstrate that the fully discrete scheme preserves kinetic energy in the inviscid limit ($\mathrm{Re} \to \infty$) and in the absence of external forcing ($\boldsymbol{f} = 0$).

5.2.1 Conservation of $\mathcal{K}_2$

Consider the integer time step equation (5.1) (with $\mathrm{Re} \to \infty$ and $\boldsymbol{f} = 0$ or $\boldsymbol{f} = \nabla\phi$). Take the L^2 inner product with $\boldsymbol{v}_2^h = (\boldsymbol{u}_2^{h,k} + \boldsymbol{u}_2^{h,k-1})/2 \in V_h^2$. Using the fact that $\boldsymbol{\omega}_2^{h,k-\frac{1}{2}} \in V_h^2$ is known from the previous half-integer step, we obtain

$$\left\langle \frac{\boldsymbol{u}_2^{h,k} - \boldsymbol{u}_2^{h,k-1}}{\Delta t}, \frac{\boldsymbol{u}_2^{h,k} + \boldsymbol{u}_2^{h,k-1}}{2} \right\rangle_\Omega + \left\langle \boldsymbol{\omega}_2^{h,k-\frac{1}{2}} \times \frac{\boldsymbol{u}_2^{h,k} + \boldsymbol{u}_2^{h,k-1}}{2}, \frac{\boldsymbol{u}_2^{h,k} + \boldsymbol{u}_2^{h,k-1}}{2} \right\rangle_\Omega = 0. \tag{5.7}$$

The second term vanishes because for any vectors $\boldsymbol{a}, \boldsymbol{b}$ we have $\langle \boldsymbol{a} \times \boldsymbol{b}, \boldsymbol{b} \rangle = 0$ (anti-symmetry of the cross product). The first term can be rewritten as

$$\frac{1}{\Delta t}\left(\frac{1}{2}\langle \boldsymbol{u}_2^{h,k}, \boldsymbol{u}_2^{h,k}\rangle_\Omega - \frac{1}{2}\langle \boldsymbol{u}_2^{h,k-1}, \boldsymbol{u}_2^{h,k-1}\rangle_\Omega\right) = \frac{\mathcal{K}_2^{h,k} - \mathcal{K}_2^{h,k-1}}{\Delta t}, \tag{5.8}$$

where we have introduced the discrete kinetic energies

$$\mathcal{K}_2^{h,k} := \frac{1}{2}\langle \boldsymbol{u}_2^{h,k}, \boldsymbol{u}_2^{h,k}\rangle_\Omega, \qquad \mathcal{K}_1^{h,k+\frac{1}{2}} := \frac{1}{2}\langle \boldsymbol{u}_1^{h,k+\frac{1}{2}}, \boldsymbol{u}_1^{h,k+\frac{1}{2}}\rangle_\Omega.$$

Hence we obtain

$$\mathcal{K}_2^{h,k} = \mathcal{K}_2^{h,k-1} \qquad \forall k. \tag{5.9}$$

5.2.2 Conservation of $\mathcal{K}_1$

A completely analogous argument is applied to the half-integer step equation (5.4). Taking the inner product with $\boldsymbol{w}_1^h = (\boldsymbol{u}_1^{h,k+\frac{1}{2}} + \boldsymbol{u}_1^{h,k-\frac{1}{2}})/2 \in V_h^1$ and using the anti-symmetry of the convective term

yields

$$\left\langle \frac{\boldsymbol{u}_1^{h,k+\frac{1}{2}} - \boldsymbol{u}_1^{h,k-\frac{1}{2}}}{\Delta t}, \frac{\boldsymbol{u}_1^{h,k+\frac{1}{2}} + \boldsymbol{u}_1^{h,k-\frac{1}{2}}}{2} \right\rangle_\Omega = 0, \tag{5.10}$$

which is exactly

$$\frac{\mathcal{K}_1^{h,k+\frac{1}{2}} - \mathcal{K}_1^{h,k-\frac{1}{2}}}{\Delta t} = 0. \tag{5.11}$$

Consequently,

$$\mathcal{K}_1^{h,k+\frac{1}{2}} = \mathcal{K}_1^{h,k-\frac{1}{2}} \qquad \forall k. \tag{5.12}$$

Thus both kinetic energies $\mathcal{K}_1$ and $\mathcal{K}_2$ are conserved in the inviscid, unforced or gradient-forced case.

5.3 Helicity conservation

We now prove that the fully discrete scheme also conserves helicity. First we define two discrete helicities that correspond to the two velocity representations.

At the integer time instant t^k, set

$$\boldsymbol{u}_1^{h,k} := \frac{\boldsymbol{u}_1^{h,k+\frac{1}{2}} + \boldsymbol{u}_1^{h,k-\frac{1}{2}}}{2}, \qquad \boldsymbol{\omega}_2^{h,k} := \frac{\boldsymbol{\omega}_2^{h,k+\frac{1}{2}} + \boldsymbol{\omega}_2^{h,k-\frac{1}{2}}}{2}, \tag{5.13}$$

and recall that the discrete vorticity fields are defined through the curl operators:

$$\boldsymbol{\omega}_1^{h,k} = \nabla_h \times \boldsymbol{u}_2^{h,k}, \qquad \boldsymbol{\omega}_2^{h,k+\frac{1}{2}} = \nabla_h \times \boldsymbol{u}_1^{h,k+\frac{1}{2}}. \tag{5.14}$$

Then we introduce

$$\mathcal{H}_1^{h,k} := \langle \boldsymbol{u}_1^{h,k}, \boldsymbol{\omega}_1^{h,k} \rangle_\Omega, \qquad \mathcal{H}_2^{h,k} := \langle \boldsymbol{u}_2^{h,k}, \boldsymbol{\omega}_2^{h,k} \rangle_\Omega. \tag{5.15}$$

5.3.1 Equality of the two helicities

Using the algebraic identity lemma 3.2, we obtain

$$\begin{aligned}
\mathcal{H}_1^{h,k} &= \frac{1}{2}\left\langle \nabla_h \times \boldsymbol{u}_2^{h,k}, \boldsymbol{u}_1^{h,k+\frac{1}{2}} + \boldsymbol{u}_1^{h,k-\frac{1}{2}} \right\rangle_\Omega \\
&= \frac{1}{2}\left(\langle \boldsymbol{u}_2^{h,k}, \nabla_h \times \boldsymbol{u}_1^{h,k+\frac{1}{2}} \rangle_\Omega + \langle \boldsymbol{u}_2^{h,k}, \nabla_h \times \boldsymbol{u}_1^{h,k-\frac{1}{2}} \rangle_\Omega \right) \\
&= \frac{1}{2}\left\langle \boldsymbol{u}_2^{h,k}, \boldsymbol{\omega}_2^{h,k+\frac{1}{2}} + \boldsymbol{\omega}_2^{h,k-\frac{1}{2}} \right\rangle_\Omega = \mathcal{H}_2^{h,k}.
\end{aligned} \tag{5.16}$$

Hence $\mathcal{H}_1^{h,k} = \mathcal{H}_2^{h,k}$ for every k; it suffices to show the conservation of one of them.

5.3.2 Conservation of $\mathcal{H}_1^{h,k}$

We again work in the inviscid, unforced setting. Start from the half-integer step equation (5.4) and take the test function $\boldsymbol{\epsilon}_1^h = \boldsymbol{\omega}_1^{h,k} \in V_h^1$ (note that $\boldsymbol{\omega}_1^{h,k}$ belongs to V_h^1 because $\nabla_h \times : V_h^2 \to V_h^1$). This gives

$$\left\langle \frac{\boldsymbol{u}_1^{h,k+\frac{1}{2}} - \boldsymbol{u}_1^{h,k-\frac{1}{2}}}{\Delta t}, \boldsymbol{\omega}_1^{h,k} \right\rangle_\Omega + \left\langle \boldsymbol{\omega}_1^{h,k} \times \frac{\boldsymbol{u}_1^{h,k+\frac{1}{2}} + \boldsymbol{u}_1^{h,k-\frac{1}{2}}}{2}, \boldsymbol{\omega}_1^{h,k} \right\rangle_\Omega = 0. \tag{5.17}$$

The second term vanishes again by anti-symmetry, yielding

$$\langle \boldsymbol{u}_1^{h,k+\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k} \rangle_\Omega = \langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k} \rangle_\Omega. \tag{A}$$

Next we employ the integer step equation (5.1) with the test function $\nabla_h \times \boldsymbol{\epsilon}_1$ for an arbitrary $\boldsymbol{\epsilon}_1^h \in V_h^1$. Thanks to lemma 3.2, we have

$$\left\langle \frac{\boldsymbol{u}_2^{h,k} - \boldsymbol{u}_2^{h,k-1}}{\Delta t}, \nabla_h \times \boldsymbol{\epsilon}_1 \right\rangle_\Omega = \left\langle \frac{\boldsymbol{\omega}_1^{h,k} - \boldsymbol{\omega}_1^{h,k-1}}{\Delta t}, \boldsymbol{\epsilon}_1 \right\rangle_\Omega.$$

Choosing $\boldsymbol{\epsilon}_1^h = \boldsymbol{u}_1^{h,k-\frac{1}{2}}$ and observing that $\nabla_h \times \boldsymbol{u}_1^{h,k-\frac{1}{2}} = \boldsymbol{\omega}_2^{h,k-\frac{1}{2}}$, the convective term becomes

$$\left\langle \boldsymbol{\omega}_2^{h,k-\frac{1}{2}} \times \frac{\boldsymbol{u}_2^{h,k} + \boldsymbol{u}_2^{h,k-1}}{2}, \nabla_h \times \boldsymbol{u}_1^{h,k-\frac{1}{2}} \right\rangle_\Omega = \left\langle \boldsymbol{\omega}_2^{h,k-\frac{1}{2}} \times \frac{\boldsymbol{u}_2^{h,k} + \boldsymbol{u}_2^{h,k-1}}{2}, \boldsymbol{\omega}_2^{h,k-\frac{1}{2}} \right\rangle_\Omega = 0.$$

Therefore

$$\left\langle \frac{\boldsymbol{\omega}_1^{h,k} - \boldsymbol{\omega}_1^{h,k-1}}{\Delta t}, \boldsymbol{u}_1^{h,k-\frac{1}{2}} \right\rangle_\Omega = 0, \tag{5.18}$$

i.e.

$$\langle \boldsymbol{\omega}_1^{h,k}, \boldsymbol{u}_1^{h,k-\frac{1}{2}} \rangle_\Omega = \langle \boldsymbol{\omega}_1^{h,k-1}, \boldsymbol{u}_1^{h,k-\frac{1}{2}} \rangle_\Omega. \tag{B}$$

Combining (A) and (B) we obtain

$$\langle \boldsymbol{u}_1^{h,k+\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k} \rangle_\Omega = \langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k-1} \rangle_\Omega. \tag{5.19}$$

Now using the definition of $\mathcal{H}_1^{h,k}$ and the midpoint averages,

$$\begin{aligned}
\mathcal{H}_1^{h,k} &= \frac{1}{2}\Big(\langle \boldsymbol{u}_1^{h,k+\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k} \rangle_\Omega + \langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k} \rangle_\Omega \Big) \\
&= \frac{1}{2}\Big(\langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k-1} \rangle_\Omega + \langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k} \rangle_\Omega \Big) \\
&= \frac{1}{2}\Big(\langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k-1} \rangle_\Omega + \langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k-1} \rangle_\Omega \Big) = \mathcal{H}_1^{h,k-1},
\end{aligned} \tag{5.20}$$

where in the last equality we used (B) again to replace $\langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k} \rangle$ by $\langle \boldsymbol{u}_1^{h,k-\frac{1}{2}}, \boldsymbol{\omega}_1^{h,k-1} \rangle$. Hence

$$\mathcal{H}_1^{h,k} = \mathcal{H}_1^{h,k-1} \qquad \forall k, \tag{5.21}$$

and because of (5.16) the same holds for $\mathcal{H}_2^{h,k}$. This proves helicity conservation in the fully discrete scheme.

5.4 Dissipation rates in the viscous case

When viscosity is present ($\mathrm{Re} < \infty$), the above inner products retain the dissipative terms. Repeating the arguments with the full equations (5.1) and (5.4) yields the following discrete dissipation rates.

5.4.1 Kinetic energy dissipation

For the integer step:

$$\frac{\mathcal{K}_2^{h,k} - \mathcal{K}_2^{h,k-1}}{\Delta t} = -\frac{1}{\mathrm{Re}} \left\| \frac{\boldsymbol{\omega}_1^{h,k} + \boldsymbol{\omega}_1^{h,k-1}}{2} \right\|_{L^2(\Omega)}^2 \leq 0. \tag{5.22}$$

For the half-integer step:

$$\frac{\mathcal{K}_1^{h,k+\frac{1}{2}} - \mathcal{K}_1^{h,k-\frac{1}{2}}}{\Delta t} = -\frac{1}{\mathrm{Re}} \left\| \frac{\boldsymbol{\omega}_2^{h,k+\frac{1}{2}} + \boldsymbol{\omega}_2^{h,k-\frac{1}{2}}}{2} \right\|_{L^2(\Omega)}^2 \leq 0. \tag{5.23}$$

These relations are the discrete counterparts of the continuous dissipation law.

5.4.2 Helicity dissipation

A lengthier but straightforward calculation (combining the two evolution equations and employing the lemma 3.2) gives

$$\frac{\mathcal{H}_1^{h,k} - \mathcal{H}_1^{h,k-1}}{\Delta t} = -\frac{2}{\mathrm{Re}} \left\langle \frac{\boldsymbol{\omega}_2^{h,k+\frac{1}{2}} + \boldsymbol{\omega}_2^{h,k-\frac{1}{2}}}{2}, \nabla_h \times \frac{\boldsymbol{\omega}_1^{h,k} + \boldsymbol{\omega}_1^{h,k-1}}{2} \right\rangle_\Omega . \tag{5.24}$$

This expression is exactly the discrete version of the continuous helicity dissipation rate; note that it may be positive or negative, reflecting the possible generation of helicity in viscous flows.

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